

Control on process parameters and Countermeasures against Process Deviation for PTA UNIT

Contents

No	Equipment	Page	No	Equipment	Page	No	Equipment	Page	No	Equipment	Page
1	PTK-101	2	17	PD-301	13	33	PJ-404	26	49	PM-500A/B	42
2	PM-104	2	18	PJ-301	14	34	PM-901	27	50	PE-502	42
3	PD-101	3	19	PD-302	15	35	PM-402A/B	28	51	PE-503	42
4	PM-105	5	20	PJ-302	16	36	PD-405	29	52	PE-504	44
5	PP-101A/B	5	21	PD-303	16	37	PC-403A(B)	30	53	PD-1206	45
6	PP-102A/B/C	6	22	PJ-303	17	38	PD-406	31	54	PE-403	46
7	PE-101A	7	23	PD-304	18	39	PM-404	31	55	PF-601	47
8	PE-101B	7	24	PJ-304	19	40	PM-404-2	35	56	PD-601	49
9	PE-101C	8	25	PM-301	19	41	PM-404-1, PZ-403	35	57	PTK-600	50
10	PE-101D	8	26	PM-401A/B/C	20	42	PC-402	36	58	PD-901	51
11	PE-101E	9	27	PD-401	22	43	PT-401	36	59	PD-603	52
12	PE-101F	9	28	PJ-401	23	44	PTK-800A/B/C	38	60	PC-601, PC-602	52
13	PD-201	9	29	PD-402	23	45	PM-803A/B/C	38	61	PE-605	52
14	PD-202	11	30	PD-403	24	46	PT-501	39	62	PD-901	53
15	PC-201A/B	12	31	PJ-402	26	47	PD-501	40	63	PJ-901	53
16	PP-201A/B	13	32	PD-404	26	48	PD-502	41	64	PM-903A/B	53

Control on process parameters and Countermeasures against Process Deviation
(for PTA UNIT)

Equipment	Factor	Operational Control Figure			Reason for Setting of Critical Figure	Method of Detection	Example of Action to be Taken upon Deviation of Control Figure from Normal One
		Normal	Allowable	Critical			
1. TA Hopper (PTK-101)	1) Powder Level (1) Upper Limit (WISK-2101)	4-8ton	a) 11ton (Alarm)	b) 13ton Interlock	(1) It is specified to prevent CTA from overflowing the hopper and thereby plugging the bag filter PM-104 due to overcharging CTA into the hopper. (Note) LA-2100 supplementing a malfunction of WISK-2101 is a fixed type level meter which is equivalent to the CTA of 16.8 tons.	Weighing Meter with Alarm and Interlock (WISK-2101)	(1)-a)-① Decrease an amount of CTA being received from the TA unit. ② Compare the indication on WISK-2101 with the actual level to be confirmed in the field.
	(LA-2100)			16.8ton (Fixed)		Fixed Powder Level Meter with Alarm (LA-2100)	(1)-b)-① By the actuation of the Interlock, TZ-400 stops and HZSV-1400 closes, with a result that CTA being received from the TA unit is stopped completely.
	(2) Lower Limit (WISK-2101)	4-8ton	2.5ton (Alarm)	2ton	(2) When PTK-101 is empty, the CTA to the reactor is stopped, thereby resulting in a decrease in the production of PTA product. Further, the powder level in PTK-101 must be kept constant in order to secure an accurate measurement of CTA on PM-101.	Weighing Meter with Alarm and Interlock (WISK-2101)	(2)-① Increase an amount of CTA being received from the TA unit. ② Check the silo, from which CTA is sent to PTA unit, to be sure that CTA in stock is being discharged smoothly.
2. Bag Filter (PM-104)	1) Differential Pressure Upper Limit (PDI-2104)	30mmAq	a) 200mmAq (Alarm)	b) 250mmAq	The filter cloth and filter retainer may be collapsed by the differential pressure of 250mmAq or higher.	Differential Pressure Indicator with Alarm (PDI-2104)	a)-① Check the indication on PDI-2104. ② Check the working condition of pulse unit and adjust the timer to acuate the pulse unit frequently, if necessary. ③ Check to be sure that there is no TA product accumulated at the bottom of PM-104. Also, check the indication on TJI-2104.

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(PM-104)							④ Check the flow rate of waste gas to be used for pneumatic transportation. b)-① Replace the filter cloth with the new one, if it is considered the end of life.
	2) <u>Pressure</u> Upper Limit (PI-2105)	200mmAq	a) 500mmAq (Alarm)	b) 800 mm Aq (PRV-2104)	It is specified to prevent the silo with design pressure of 0.1 kg/cm ² G from being damaged due to overpressure, though the bag filter is designed by the pressure of 0.13 kg/cm ² G.	Pressure Indicator with Alarm (PI-2105)	a)-① Decrease an amount of CTA received from TA unit and the waste gas to be used for transporting CTA to PTA unit. ② Confirm that the pulse unit is functioning properly, and check the differential pressure on PDI-2104 to see if the bag filter is not clogged with TA. ③ Check to be sure that an excess amount of N ₂ is not introduced through the aerators installed at the bottom of the bag filter. b)-① Confirm that PRV-2104 is functioning properly.
3. Feed Slurry Drum (PD-101)	1) <u>Liquid Level</u> (1) Upper Limit (LIK-2101)	60%	75% (Alarm)	80%	It is specified to prevent a liquid from overflowing from PD-101. (Note) The critical figure is decided on the figure which can be confirmed through the SG provided on the upper part of the drum.	Level Indicator with Alarm (LICK-2101)	(1)-① Compare the indication of LICK-2101 with that on the LG-2101 and the actual level through the SG. ② Check the indications on FICQK-2101, FICK-2102, FI-2201 and FIC-2201A,B(C), and adjust their flow rates, if necessary. ③ Check the slurry density indicated on AI-2100 and AI-2101. ④ Decrease an amount of CTA fed to PD-101 or stop feeding CTA to PD-101 once to reduce the level to the normal one.

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(PD-101)	(2)Lower Limit (LIK-2101)	60%	40% (Alarm)	35%	(2)It is specified to mix CTA with the recycle solvent effectively as well as to prevent TP-101 from getting into the cavitation phenomenon.		(2)-① Take the same actions as mentioned in the items (1)-①, ② and ③ above. ② Raise the level in PD-101 by temporarily increasing the recycle solvent on FICK-2102.
	2) <u>Density Meter</u> (1)Upper Limit (AI-2100 or AI-2101)	1.076	1.081 (Alarm)	1.125 (Alarm)	(1)The reactor temperature is decided so as to have an allowance of 15% against the saturated solubility of TA into water based on the slurry fed to the PTA reactor. And so, as the slurry concentration in PD-101 increases, there may be a possibility that a TA dissolved in water crystallizes in the reactor, thereby crushing the pd/c catalyst.	Density Indicator with Alarm (AI-2100)	(1)-① When the critical alarm sounds, immediately decrease an amount of CTA fed to PD-101 because at the said condition, the temperature allowance to the precipitating point lowers to about 3°C from 5.6°C. ② Purge the line having the density meter installed with LPW and check to be sure that the indication on AI-2100 or AI-2101 is the reliable figure. ③ Check the flow rates of CTA and solvent fed to PD-101 through FICQK-2101 and FICK-2102 respectively, and adjust them, if necessary. ④ Reset the set-point on TIC-2101 to match an increase of slurry concentration in PD-101.
	(2)Lower Limit	1.076	1.071 (Alarm)	1.0750 (Alarm)	(2)It is specified to quickly detect the trouble associated with the TA feeder and the flow controller for the recycle solvent.		(2)-① Check the flow rate of CTA and R.S. fed to PD-101 (FICQK-2101, FICK-2102) ② Purge AI-2100 or AI-2101 with LPW and check the reliability of the indication on AI-2100 or AI-2101.

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(PD-101)	3) <u>Recycle Solvent</u> Lower Limit (FICK-2102)	78.5	76 (Alarm)	74	If the flow rate of the solvent fed to PD-101 is insufficient, the concentration of solid increases, with a result that there is a fear of plugging pd/c catalyst with a TA crystallized in the PTA reactor.	Flow Indicator and Regulator with Alarm (FICK-2102)	① Check the indications on FICQK-2101, AI-2100 or AI-2101, FICK-2102, FICK-2103, FICK-2104 and FICK-2105. ② Check the working condition of PP-501 and the level of PD-501 on LIC-2501A and B. ③ After checking the flow rate on FICK-2102, adjust the flow rate of the solvent fed to PD-101, if necessary. ④ Check the slurry density indicated on AI-2100 or AI-2101.
4. Flow Jet Mixer (PM-105)	<u>Recycle Solvent</u> Lower Limit (FICK-2105)	20t/hr	17t/hr (Alarm)	15t/hr	PM-105 serves to prepare the slurry of about 60% mixing the TA with the solvent before entering PD-101. And so, a shortage of an amount of solvent may causes a plugging at the outlet of PM-105 and a damage of the equipment.	Flow Indicator and Regulator with Alarm (FICK-2105)	① Confirm that FICK-2105 is working properly. ② Check the indication on FICK-2105 and adjust the flow rate, if necessary. ③ Check the premixing condition made by PM-105 through the S.G. on PD-101.
5. Feed Booster Pump (PP-101 A/B)	<u>Pressure</u> (1) Lower Limit (PIC-2101)	11.5 kg/cm ² G	10 kg/cm ² G (Alarm)	9 kg/cm ² G	(1) PP-101 is the pump to feed slurry to the REACTOR FEED PUMP PP-102. With the delivery pressure kept constant, it keeps stable feed to PP-102 and prevents PP-102 from getting into the cavitation.	Pressure Indicator and Regulator with Alarm (PIC-2101)	(1)-① Check the indication on PIC-2101. ② Purge the line having PIC-2101 installed with LPW. ③ Switch PP-101 over to the stand-by side and clean the strainer installed on the suction of PP-101. ④ Check the level of PD-101.

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(PP-101 A/B)	(2)Upper Limit (PIC-2101)	11.5 kg/cm ² G	12.5 kg/cm ² G (Alarm)	13.5 kg/cm ² G	It is specified to prevent the sealant to the mechanical seal of PP-102 from being obstructed with a rise of pressure.	Ditto	(2)-① Check the indication on PIC-2101 ② Adjust the by-pass valve of PV-2101, if necessary.
6.Reactor Feed Pump (PP-102 A/B/C)	1)Ampere Upper Limit (II-2102A/B/C)	46-48A	51.5A (Alarm)	54A (Rated)	It is specified to prevent the motor coil from damaging due to the overload.	Ampere Indicator with Alarm (II-2102A/B/C)	(1)-① Check to be sure that the flow rate of slurry fed to the reactor is an appropriate one. ② Switch PP-102 over to the stand-by side and check it.
	2)Pressure (1)Upper Limit (PI-2103)	90-92 kg/cm ² G	95 kg/cm ² G (Alarm)	120 kg/cm ² G SP for SV	(1)It is specified by taking into consideration the equipment for the feed and reactor systems designed by the pressure of 120 kg/cm ² G.	Pressure Indicator with Alarm (PI-2103)	(1)-① Check to see if there is no plugging between PE-101A and PE-101F by comparing the indication on PI-2203 with that on PI-2103. ② Check the pressure on the outlet of PP-101. (PIC-2101) ③ Check the rate of solution fed to the reactor. ④ Stop feeding CTA to PD-101, and switch the operation over to the Recycle Solvent Run.
	(2)Lower Limit (PI-2103)	90-92 kg/cm ² G	85 kg/cm ² G (Alarm)	80 kg/cm ² G	(2)It is specified to detect the abnormal condition relating to the Sundyne Pump at an earlier stage.	Ditto	(2)-① Check the pressure on the outlet of PP-101 and switch PP-101 over to the stand-by side, if necessary. ② Check the flow rate of TA solution fed to the PTA reactor. ③ Switch PP-102 over to the stand-by side, if necessary.

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7. Feed Preheater (PE-101A)	<u>Temperature</u> (1) Upper Limit (TIC-2105)	122°C	140°C (Alarm)	145°C (Design)	It is specified to prevent the temperature of PE-101A from rising over the design condition by use of steam generated in the crystallizers PD-304 and steam condensate in the preheater PE-101B.	Temperature Indicator with Alarm	(1)-① Compare the indication on TIC-2105. ② Check the working condition of LV-2103. ③ Check the working condition of TV-2105, and shut the block valve with TV-2105, if 10K steam is leaking into the PE-101A side.
	(2) Lower Limit (TIC-2105)	122°C	120°C (Alarm)	120°C	It is specified to detect the fouling of PE-101A shell side.		(2)-① Compare the indication on TIC-2105. ② Check the working condition of LIC-2103 and PIC-2304, and the valve opening degrees for LV-2103 and PV-2304B. ③ Confirm that LIC-2102 installed on the SC drain pot is functioning properly.
8. Feed Preheater (PE-101B)	<u>Temperature</u> (1) Upper Limit (TIC-2106)	156°C	175°C (Alarm)	180°C (Design)	It is specified to prevent the temperature of PE-101B from rising over the design condition by use of steam generated in the crystallizers PD-303 and steam condensate in the preheater PE-101C.	Temperature Indicator with Alarm	(1)-① Compare the indication on TIC-2106. ② Check the working condition of LV-2104. ③ Check the working condition of TV-2106, and shut the block valve with TV-2106, if 10K steam is leaking into the PE-101B side. ④ Check the working condition of FV-2302.
	(2) Lower Limit (TIC-2106)	156°C	154°C (Alarm)	154°C	It is specified to detect the fouling of PE-101B shell side.		(2)-① Compare the indication on TIC-2106. ② Check the working condition of LIC-2104 and PIC-2303, and the valve opening degrees for LV-2104 and PV-2303B. ③ Confirm that LIC-2103 installed on the SC drain pot is functioning properly. ④ Check the same action as above (1)-④.

MPS-07012

Control on Process Parameters

and

Countermeasures against Process Deviation

(for PTA UNIT)

Preliminary information
reference only

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9. Feed Preheater (PE-101C)	<u>Temperature</u> (1)Upper Limit (TIC-2107)	194°C	210°C (Alarm)	215°C (Design)	It is specified to prevent the temperature of PE-101C from rising over the design condition by use of steam generated in the crystallizers PD-302 and steam condensate in the preheater PE-101D.	Temperature Indicator with Alarm	(1)-① Compare the indication on TIC-2107. ② Check the working condition of LV-2105. ③ Check the working condition of TV-2107, and shut the block valve with TV-2107, if 10K steam is leaking into the PE-101C side.
	(2)Lower Limit (TIC-2107)	194°C	192°C (Alarm)	192°C	It is specified to detect the fouling of PE-101C shell side.		(2)-① Compare the indication on TIC-2107. ② Check the working condition of LIC-2105 and PIC-2302, and the valve opening degrees for LV-2105 and PV-2302B. ③ Confirm that LIC-2104 installed on the SC drain pot is functioning properly.
10. Feed Preheater (PE-101D)	<u>Temperature</u> (1)Upper Limit (TJI-2108)	245°C	260°C (Alarm)	265°C (Design)	It is specified to prevent the temperature of PE-101D from rising over the design condition by use of steam generated in the crystallizers PD-301.	Temperature Indicator with Alarm	(1)-① Compare the indication on TJI-2108. ② Check the working condition of PV-2301A and B, and indication on PIC-2301.
	(2)Lower Limit (TJI-2108)	245°C	243°C (Alarm)	243°C	It is specified to detect the fouling of PE-101D shell side.		(2)-① Compare the indication on TJI-2108. ② Check the working condition of PIC-2301, and the valve opening degrees for PV-2301B. ③ Confirm that LIC-2105 installed on the SC drain pot is functioning properly.

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11. Feed Preheater (PE-101E)	Temperature (1)Upper Limit (TJI-2109)	264°C	290°C (Alarm)	295°C (Design)	It is specified to prevent the temperature of PE-101E from rising over the design condition by use of hot oil heated in the hot oil heater.	Temperature Indicator with Alarm	(1)-① Compare the indication on TJI-2109. ② Check the working condition of TV-2101. ③ Check the indication on FI-2604. ④ Check the indication on TJI-2619 and TIC-2601.
	(2)Lower Limit (TJI-2109)	264°C	262°C (Alarm)	262°C	It is specified to detect the fouling of PE-101E shell side.		(2)-① Check the same action as above (1)-① ~④.
12. Feed Preheater (PE-101F)	Temperature (1)Upper Limit (TIC-2101)	287°C	330°C (Alarm)	335°C (Design)	It is specified to prevent the temperature of PE-101F from rising over the design condition by use of hot oil heated in the hot oil heater.	Temperature Indicator with Alarm	(1)-① Compare the indication on TIC-2101. ② Check the working condition of TV-2101. ③ Check the indication on FI-2604. ④ Check the indication on TJI-2619 and TIC-2601.
	(2)Lower Limit (TIC-2101)	287°C	285°C (Alarm)	285°C	It is specified to detect the fouling of PE-101F shell side.		(2)-① Check the same action as above (1)-① ~④.
13. Reactor (PD-201)	(1)Flow Rate Lower Limit (FI-2201 or FIC-2201A,B(C))	151.8 t/hr	119t/hr (Alarm)	85t/hr	The Sundyne pump having a special Head-Flow performance curve can not develop the expected performance under the specified flow rate. Note; PP-102 Performance Head 900m Minimum Flow 40m³/hr	Flow Indicator and Regulator with Alarm (FI-2201 or FIC-2201A,B(C))	① Confirm the reliability of the indication on FI-2201 and FIC-2201A,B(C) by checking the amperage of PP-102 and the variation of the level of PD-101. ② Based on the differential pressure between the indications on PI-2103 and PI-2203, and the opening degree of FIC-2201A,B(C), check to see if a fall of flow on FIC-2201A,B(C) is caused by a plugging of tubes in the PE-101A through PE-101F.

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(PD-201)							③ In case where the flow rate can't be returned to the normal one due to the reason said above, switch the operation over to the Hot Water Run.
	2) <u>Pressure</u> Upper Limit (PICK-2201)	77.5 kg/cm ² G	a) 80 kg/cm ² G (Alarm)	b) 85 kg/cm ² G Interlock	It is specified to detect the clogging of the reactor with TA crystallized. (Note) The critical figure is based on the pressure which doesn't cause the breakage of the rupture disc by mistake.	Pressure Indicator and Regulator with Alarm (PICK-2201)	a) -① Confirm that PICK-2201 is functioning properly by comparing the indication on PICK-2201 with those on PI-2203 and PG-2203. ② Check the slurry density on AI-2100 or AI-2101 and the reactor temperature on TIK-2201. b) -① If the pressure rises to 85 kg/cm ² G, the operation is changed over to the Recycle Solvent Operation by the actuation of the Interlock. ② In case where the pressure rises even with the action taken in the item ① above, stop PP-102 and execute ESD.
	3) <u>Liquid Level</u> Upper Limit (LICK-2201)	50%	75% (Alarm)	97% (Max)	It is specified to detect the clogging of the reactor with TA or the failure of H ₂ system.	Level Indicator and Regulator with Alarm (LICK-2201)	① Check the opening degree of PICK-2201 and the indications on TIC-2101, FICK-2202 and PIC-2202. ② Confirm that H ₂ compressor PC-201 is functioning properly. If H ₂ flow rate decreases due to a malfunction of PC-201, switch it over to the stand-by side. ③ If the level is over 100% and the pressure rises to 85 kg/cm ² G, the operation is changed over to the Recycle Solvent Run by the actuation of the Interlock.

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(PD-201)	4) <u>Temperature</u> Lower Limit (TIK-2201)	286.6°C	285°C (Alarm)	283°C	Reactor temperature is determined by the slurry concentration in TA solution fed to PD-201. If the reactor temperature is lower than the temperature at which water is saturated with the dissolved TA, the catalyst layer fixed in the reactor is clogged with TA crystallized.	Temperature Indicator with Alarm (TIK-2201)	① Compare the indication on TIK-2201 with that on TJI-2200 and check to be sure that TIK-2201 is functioning properly. ② Check the outlet temperature of each Feed Preheater and confirm that each Preheater is not having trouble with heating source. ③ Check to see if Hot Oil Heater PF-601 is functioning properly. ④ If the temperature on TIK-2201 can't be adjusted due to a malfunction of TIC-2101, adjust it by controlling TIC-2101 on manual while checking the flow rate of HO fed to PE-101E & F on FI-2604. ⑤ If the temperature on TIK-2201 can't be adjusted to the specified figure even with the action taken in the item ④ above, change the operation over to the Recycle Solvent Operation.
14. Hydrogen Buffer Drum (PD-202)	1) <u>Liquid Level</u> Upper Limit (LA-2202)			Fixed Point (Alarm)	It is specified to prevent liquid trapped in PD-202 from being fed to PD-201, which is designed to compress the H ₂ gas.	Fixed Type Liquid Level Indicator with Alarm (LA-2202)	① Check the actual level on LG-2202 in the field. ② Blow down the drain trapped in PD-202 through the drain valve.
	2) <u>Temperature</u> (1) Upper Limit (TJI-2202)	20~50°C	55°C (Alarm)	60°C	(1) It is specified by taking into consideration the design temperature of 60°C for PD-202.	Temperature Indicator with Alarm (TJI-2202)	(1)-① Check the flow rate of CCW fed to the PLC-201 and adjust it, if necessary. ② Spray water on PD-202 to cool it down to the specified temperature.

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(PD-202)	(2) Lower Limit (TJI-2202)	20~50°C	10°C (Alarm)	2°C	(2) It is specified to prevent water condensed in H ₂ gas from clogging the line with it.	Ditto	
15. H ₂ Compressor (PC-201 A/B)	1) <u>Diaphragm Pressure</u> Upper Limit (PA-2204A/B)	0 kg/cm ² G		0.5 kg/cm ² G (Alarm)	If the diaphragm is damaged, the compressor doesn't work properly.	Pressure Switch with Alarm (PA-2204A/B)	① Switch PC-201 over to the stand-by side and replace the diaphragm with new one.
	2) <u>Oil Pressure</u> Lower Limit (PA-2205A/B)	5.0 kg/cm ² G		2.0 kg/cm ² G (Alarm)	If the oil pressure falls, it may lead to the damage of the bearing and a fall of flow rate of H ₂ discharged from the H ₂ compressor.	Pressure Switch with Alarm (PA-2205A/B)	① Switch PC-201 over to the stand-by side and repair it.
	3) <u>Delivery Pressure</u> (1) Upper Limit (PIC-2202)	90 kg/cm ² G	100 kg/cm ² G (Alarm)	108 kg/cm ² G (Design)	It is specified to prevent the compressor from being pressurized over the design pressure.	Pressure Indicator and Regulator with Alarm (PIC-2202)	(1)-① Compare the indication on PIC-2202 with that on PG-2208 to confirm that the indication on PIC-2202 is correct. ② Confirm that PIC-2202 is functioning properly. ③ Adjust H ₂ pressure by controlling PIC-2202 on manual.
	(2) Lower Limit (PIC-2202)	90 kg/cm ² G	86 kg/cm ² G (Alarm)	85 kg/cm ² G	(2) If the pressure falls, the control of the level in PD-201 becomes difficult.	Ditto	(2)-① Take the actions as mentioned in the items (1)- ①, ② & ③ above. ② Switch PC-201 over to the stand-by side. ③ Check the pressure on the suction of PC-201 on PG-2206. ④ Confirm that PICSK-2201 and LICK-2201 are functioning properly.

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(PC-201 A/B)							⑤ If the H ₂ pressure on PIC-2202 can not be returned to the normal figure even with the actions above, change the operation over to the Recycle Solvent Run.
16. High Pressure Flush Water Pump (PP-201 A/B)	Flow Rate Lower Limit (FI-2700)	500kg/hr	400 kg/hr	200 kg/hr	It is specified to prevent PCV-2201 from getting into a malfunction.	Flow Indicator with Alarm (FI-2700)	① Check the leakage through the gland seal of PV-2201, and tighten it, if any leakage. ② Check the pressure on PIC-2502 to be sure that it is operated under the specified pressure. ③ Switch PP-201 over to the stand-by side.
17. Crystal-lizer (PD-301)	1) Pressure Upper Limit (PIC-2301)	41.1 kg/cm ² G	44 kg/cm ² G	49 kg/cm ² G	It is specified to prevent PD-301 from being overpressurized. (Note) The critical figure is based on the pressure which doesn't cause the breakage of the rupture disc by mistake.	Pressure Indicator and Regulator with Alarm (PIC-2301)	① Confirm that PIC-2301 is functioning properly by comparing the indication on PIC-2301 with that on PG-2301. ② Check the openings of PV-2301A/B to see if shell of PE-101D have been clogged. ③ Check PICSK-2201 to see if the flow rate of TA solution fed to PD-301 is proper. ④ If the pressure of PD-301 can't be controlled by PIC-2301, proceed to ESD.

Equipment	Factor	Operational Control Figure			Reason for Setting of Critical Figure	Method of Detection	Example of Action to be Taken upon Deviation of Control Figure from Normal One
		Normal	Allowable	Critical			
(PD-301)	2) <u>Liquid Level</u> Upper Limit (LICK-2301)	35%	45% (Alarm)	69%	As the liquid level rises due to a clogging of PD-301 discharge line, an amount of entrainment of TA solution increases with a result that the tubes of PE-101D and PV-2301A/B tend to become clogged.	Level Indicator and Regulator with Alarm (LICK-2301)	① Confirm that LICK-2301 is functioning properly by comparing the indication on LICK-2301 with that on LG-2301. ② Based on the indications on PIC-2302 and LICK-2302, confirm that TA slurry is being discharged smoothly from PD-301. If there is any symptom of clogging tendency, purge the said discharge line with MPW. ③ Switch the operation over to the Recycle Solvent Operation at the level of 69% on LICK-2301. ④ Stop PP-102 at the level of 95% on LICK-2301.
18. Agitator (PJ-301)	1) <u>Seal Pressure</u> Lower Limit (PICS-2731)	46 kg/cm ² G	44 kg/cm ² G (Alarm)	43 kg/cm ² G Interlock	The mechanical seal of PJ-301 is pressurized to the pressure 5 kg/cm ² G higher than the operating pressure of PD-301. From view point of the fabrication for the mechanical seal, a reverse pressure on it may damage the seal part.	Pressure Indicator and Regulator with Alarm and Interlock (PICS-2731)	① Upon receiving the alarm sound, check the operating condition of PP-504 and confirm that PICS-2731 is functioning properly. ② At the pressure of 43 kg/cm ² G on PICS-2731, SDV-2731 is closed by the actuation of the Interlock.
	2) <u>Ampere</u> Upper Limit (II-2301)	48A	65A (Alarm)	73A (Rated)	It is specified to prevent the motor from being damaged due to the overload.	Ampere Indicator with Alarm (II-2301)	① Check the indication on LICK-2301. ② Check to be sure that there is not abnormality with the mechanical seal of PJ-301. ③ Check to be sure that there is not abnormality with the motor and the reducer. ④ If it is judged to be impossible of continuing the operation based on the results as checked in the items ①, ② and ③ above, stop PJ-301 after switching the operation over to the Recycle Solvent Run.

Equipment	Factor	Operational Control Figure			Reason for Setting of Critical Figure	Method of Detection	Example of Action to be Taken upon Deviation of Control Figure from Normal One
		Normal	Allowable	Critical			
19.Crystal-lizer (PD-302)	1) <u>Pressure</u> Upper Limit (PIC-2302)	16.6 kg/cm ² G	19.0 kg/cm ² G (Alarm)	23.0 kg/cm ² G	It is specified to prevent PD-302 from being overpressurized. (Note)The critical figure is based on the pressure which doesn't cause the breakage of the rupture disc by mistake.	Pressure Indicator and Regulator with Alarm (PIC-2302)	① Check the indications on LICK-2301 and PIC-2301 to be sure that an abnormal amount of TA solution is not being discharged to PD-302 from PD-301. ② Confirm that PIC-2302 is functioning properly by comparing the indication on PIC-2302 with that on PG-2302. ③ Check the openings of PV-2302A/B to see if tubes of PE-302 or the shell of PE-101C have been plugged. ④ Check the indication on FIC-2302 to be sure that vapor is condensed properly by PE-302. ⑤ If the pressure of PD-302 can't be controlled by PIC-2302, switch the operation over to the Recycle Solvent Run, by stopping feeding CTA to PD-101.
	2) <u>Liquid Level</u> Upper Limit (LICK-2302)	25%	40%	80%	As the liquid level rises due to a clogging of PD-302 discharge line, an amount of entrainment of TA solution increases with a result that the tubes of PE-302 or the shell of PE-101C and PV-2302A/B tend to become clogged.	Level Indicator and Regulator with Alarm (LICK-2302)	① Based on the indications on PIC-2303 and LICK-2303, confirm that TA slurry is being discharged smoothly from PD-302. If there is any symptom of clogging tendency, purge the said discharge line with MPW. ② Check the indications on LICK-2301 and PIC-2301 to be sure that an abnormal amount of TA solution is not being discharged to PD-302 from PD-301. ③ Confirm that LICK-2302 is functioning properly by comparing the indication on LICK-2302 with that on LG-2302. ④ Switch the operation over to the Recycle Solvent Operation at the level of 69% on LICK-2301.

Equipment	Factor	Operational Control Figure			Reason for Setting of Critical Figure	Method of Detection	Example of Action to be Taken upon Deviation of Control Figure from Normal One
		Normal	Allowable	Critical			
(PD-302)							⑤ Stop PP-102 at the level of 95% on LICK-2301 to put the Feed, Reactor and Crystallizer systems into ESD.
20. Agitator (PJ-302)	1) <u>Seal Pressure</u> Lower Limit (PICS-2732)	22 kg/cm ² G	20 kg/cm ² G (Alarm)	19 kg/cm ² G Interlock	It is specified due to the same reason as mentioned in the item 13. (PJ-301)	Pressure Indicator and Regulator with Alarm and Interlock (PICS-2732)	① Upon receiving the alarm sound, check the operating condition of PP-504 and confirm that PICS-2732 is functioning properly. ② At the pressure of 19 kg/cm ² G on PICS-2732, SDV-2732 is closed by the actuation of the Interlock.
	2) <u>Ampere</u> Upper Limit (II-2302)	34A	54A (Alarm)	60A (Rated)	It is specified to prevent the motor from being damaged due to the overload.	Ampere Indicator with Alarm (II-2302)	① Check the indication on LICK-2302. ② Check to be sure that there is not abnormality with the mechanical seal of PJ-302. ③ Take the same actions as mentioned in the items ③ and ④ for PJ-301.
21. Crystal-lizer (PD-303)	1) <u>Pressure</u> Upper Limit (PIC-2303)	8.8 kg/cm ² G	9.5 kg/cm ² G (Alarm)	10.5 kg/cm ² G	It is specified to prevent PD-303 from being overpressurized. (Note) The critical figure is based on the pressure which doesn't cause the breakage of the rupture disc by mistake.	Pressure Indicator and Regulator with Alarm (PIC-2303)	① Check the indication on LICK-2302 and PIC-2302 to be sure that an abnormal amount of TA solution is not being discharged to PD-303 from PD-302. ② Confirm that PIC-2303 is functioning properly by comparing the indication on PIC-2303 with that on PG-2303. ③ Check the openings of PV-2303A/B to see if shell of PE-101B have been plugged. ④ If the pressure of PD-303 can't be controlled by PIC-2303, switch the operation over to the Recycle Solvent Run.

Equipment	Factor	Operational Control Figure			Reason for Setting of Critical Figure	Method of Detection	Example of Action to be Taken upon Deviation of Control Figure from Normal One
		Normal	Allowable	Critical			
(PD-303)	2) <u>Liquid Level</u> Upper Limit (LICK-2303)	25%	38% (Alarm)	80%	As the liquid level rises due to a clogging of PD-303 discharge line, an amount of entrainment of TA solution increases with a result that the shell of PE-101B and PV-2303A/B tend to become clogged.	Level Indicator and Regulator with Alarm (LICK-2303)	① Based on the indications on PIC-2304 and LICK-2304, confirm that TA solution is being discharged smoothly from PD-303. If there is any symptom of clogging tendency, purge the said discharge line with MPW. ② Check the indications on LICK-2302 and PIC-2302 to be sure that an abnormal amount of TA solution is not being discharged to PD-303 from PD-302. ③ Confirm that LICK-2303 is functioning properly by comparing the indication on LICK-2303 with that on LG-2303. ④ If the level of crystallizers PD-301 to PD-303 rises to <u>69%</u> , 80% and 80% respectively due to the clogging of PD-303 discharge line, switch the operation over to the Recycle Solvent Run. ⑤ Further, if the level of PD-301 rises to <u>95%</u> , stop PP-102 to place the unit into ESD.
22. Agitator (PJ-303)	1) <u>Seal Pressure</u> Lower Limit (PICS-2733)	14.0 kg/cm ² G	11.0 kg/cm ² G (Alarm)	10.0 kg/cm ² G Interlock	It is specified due to the same reason as mentioned in the item 13. (PJ- 301) 303	Pressure Indicator and Regulator with Alarm and Interlock (PICS-2733)	① Upon receiving the alarm sound, check the operating condition of PP-504 and confirm that PICS-2733 is functioning properly. ② At the pressure of 10 kg/cm ² G on PICS-2733, SDV-2733 is closed by the actuation of the Interlock.

Equipment	Factor	Operational Control Figure			Reason for Setting of Critical Figure	Method of Detection	Example of Action to be Taken upon Deviation of Control Figure from Normal One
		Normal	Allowable	Critical			
(PJ-303)	2) <u>Ampere</u> Upper Limit (II-2303)	30A	44A (Alarm)	48.8A (Rated)	It is specified to prevent the motor from being damaged due to the overload.	Ampere Indicator with Alarm (II-2303)	① Check the indication on LICK-2303. ② Check to be sure that there is not abnormality with the mechanical seal of PJ-303. ③ Take the same actions as mentioned in the items ③ and ④ for PJ-301.
23. Crystal-lizer (PD-304)	1) <u>Pressure</u> Upper Limit (PIC-2304)	4.0 kg/cm ² G	5.0 kg/cm ² G (Alarm)	6.0 kg/cm ² G	It is specified to prevent PD-304 from being overpressurized. (Note) The critical figure is based on the pressure which doesn't cause the breakage of the rupture disc by mistake.	Pressure Indicator and Regulator with Alarm (PIC-2304)	① Check the indications on LICK-2303 and PIC-2303 to be sure that an abnormal amount of TA solution is not being discharged to PD-304 from PD-303. ② Confirm that PIC-2304 is functioning properly by comparing the indication on PIC-2304 with that on PG-2304. ③ Check the working condition of PIC-2304A/B. ④ If the pressure of PD-304 can't be controlled by PIC-2304, switch the operation over to the Recycle Solvent Run.
	2) <u>Liquid Level</u> Upper Limit (LICK-2304)	25%	40% (Alarm)	80%	As the liquid level of PD-304 rises, an amount of entrainment of TA solution increases with a result that the shell of PE-101A and PV-2304A/B tend to become clogged.	Level Indicator and Regulator with Alarm (LICK-2304)	① Check the flow rate of slurry fed to PM-401 from PD-304, and purge the feed line with LPW, if any problem. ② Check the indications on LICK-2303 and PIC-2303 to be sure that an abnormal amount of TA solution is not being discharged to PD-304 from PD-303. ③ Confirm that LICK-2304 is functioning properly by comparing the indication on LICK-2304 with that on LG-2304. ④ If the level of crystallizers PD-301 to PD-304 rises to 69%, 80%, 80% and 80% respectively due to the disorder of PD-304 discharge, switch the operation over to the Recycle Solvent Run.

Equipment	Factor	Operational Control Figure			Reason for Setting of Critical Figure	Method of Detection	Example of Action to be Taken upon Deviation of Control Figure from Normal One
		Normal	Allowable	Critical			
(PD-304)							⑤ If the level of PD-301 further rises to 95%, stop PP-102 to place the unit into ESD.
24. Agitator (PJ-304)	1) <u>Seal Pressure</u> Lower Limit (PICS-2734)	9 kg/cm ² G	7.0 kg/cm ² G (Alarm)	6.0 kg/cm ² G Interlock	It is specified due to the same reason as mentioned in the item 13. (PJ-301)	Pressure Indicator and Regulator with Alarm and Interlock (PICS-2734)	① Upon receiving the alarm sound, check the operating condition of PP-504 and confirm that PICS-2734 is functioning properly. ② At the pressure of 6.0 kg/cm ² G on PICS-2734, SDV-2734 is closed by the actuation of the Interlock.
	2) <u>Ampere</u> Upper Limit (II-2304)	30A	44A (Alarm)	48.8A (Rated)	It is specified to prevent the motor from being damaged due to the overload.	Ampere Indicator with Alarm (II-2304)	① Check the indication on LICK-2304. ② Check to be sure that there is not abnormality with the mechanical seal of PJ-304. ③ Take the same actions as mentioned in the items ③ and ④ for PJ-301.
25. Mist Separator (PM-301)	<u>Temperature</u> Lower Limit (TJI-2311)	252°C	243°C (Alarm)	242°C	PM-301 is installed to separate solid entrained with vapor vaporized from PD-301. Consequently, if the bottom of PM-301 is plugged, solid entrained with vapor is accumulated on the shell of PE-101D and leads to plug the shell. And so, it is specified to promptly detect the said plugging.	Temperature Indicator with Alarm (TJI-2311)	① Confirm that the MPW purge is introduced continuously through the restriction orifice RO-2305. ② Upon receiving the alarm sound, purge the bottom of PM-301 with MPW (Refer to the Various Works for the detail procedures). ③ If the plugging of PM-301 can not be solved even with the action above, try to purge the line by using hydraulic pump so as not to pressurize it over the design pressure.

Equipment	Factor	Operational Control Figure			Reason for Setting of Critical Figure	Method of Detection	Example of Action to be Taken upon Deviation of Control Figure from Normal One
		Normal	Allowable	Critical			
26. Pressure Centrifuge (PM-401 A/B/C)	1) <u>Pressure</u> Upper Limit (PICK-2401)	4.2 kg/cm ² G	4.6 kg/cm ² G (Alarm)	7.0 kg/cm ² G (Design)	It is specified to prevent PM-401 from being overpressurized.	Pressure Indicator and Regulator with Alarm (PICK-2401)	① Confirm that PICK-2401 is functioning properly by comparing the indication on PICK-2401 with those on PIK-2400A/B/C and PG-2413A/B/C. ② Check the working condition of PV-2401A/B and adjust the pressure by controlling PICK-2401 on manual, if necessary. ③ In case where the pressure can't be controlled even with the actions mentioned above, stop feeding slurry to PM-401 from PP-304 and depressurize PD-401 system through the drain valves for PG-2411 and PG-2413 A/B/C.
	2) <u>Torque</u> Upper Limit (XJI-2401 A/B/C)	75-110 kg·m	130 kg·m (Alarm)	172 kg·m Interlock	It is specified to prevent the conveyor and gear box of the centrifuge from being damaged due to overload.	Torque Indicator With Alarm (XJI-2401A/B/C) Torque Limitter With Interlock (ZS-2401A/B/C)	① Check the ampere on IIK-2401A/B/C and the vibration on VI-2401A/B/C and purge PM-401 with LPW, if any problem. ② When the torque value reaches 170 kg·m or larger, the torque arm trips the machine to stop feeding slurry to PM-401. ③ Switch PM-401 over to the stand-by side.
	3) <u>LO Pressure</u> Lower Limit (1) Gear Oil (PIS-3503 A/B/C)	9.4 kg/cm ² G	8.0 kg/cm ² G (Alarm)	5.0 kg/cm ² G Interlock	They are specified to prevent the centrifuge from being damaged due to a shortage of lube oil.	Pressure Indicator With Alarm (PIS-3503A/B/C) (PIS-3504A/B/C) (PDIS-3502A/B/C)	① Check the oil pressure in the field. ② Adjust the oil pressure by the pressure regulator. ③ Adjust each oil flow rate by the flow controller.

Equipment	Factor	Operational Control Figure			Reason for Setting of Critical Figure	Method of Detection	Example of Action to be Taken upon Deviation of Control Figure from Normal One
		Normal	Allowable	Critical			
(PM-401 A/B/C)	(2) Spindle Oil (PIS-3504 A/B/C)	1.2 kg/cm ² G	1.0 kg/cm ² G (Alarm)	0.5 kg/cm ² G Interlock			④ Switch the strainer installed on the outlet of L0 pump over to the stand-by side. ⑤ Switch PM-401 over to the stand-by side. ⑥ When the L0 pressure falls down to the critical figure, PM-401 is stopped by the actuation of the Interlock.
	(3) Seal Coolant (PDIS-3502 A/B/C)	1.0 kg/cm ² G	0.8 kg/cm ² G (Alarm)	0.5 kg/cm ² G Interlock			
	4) Oil Level (Oil Leakage) Lower Limit (LI-3501A/B/C)	60~80%	60% (Alarm)	30%	It is specified to prevent the damage of machine as well as the degradation of product due to a leakage of lube oil	Level Indicator With Alarm (LI-3501A/B/C)	① Confirm the oil level in the L0 tank in the field. ② In case where the oil leakage is 4 // 8hrs or more, switch PM-401 over to the stand-by side, and repair it.
	5) Vibration Upper Limit (VI-2401A/B/C)	30~150 μ	150 μ (Alarm)	200 μ	It is specified to prevent the machine from damaging based on the vibration value of 200 μ guaranteed by the vendor.	Vibration Indicator With Alarm (VI-2401A/B/C)	① Purge the centrifuge with LPW. ② When the Vibration reaches 200 μ or larger, switch PM-401 over to the stand-by side and repair it.
	6) Ampere Upper Limit (IIK-2401 A/B/C)	30~33A	35A (Alarm)	38A (Rated)	It is specified to prevent the burning of motor coil due to overcurrent.	Ampere Indicator With Alarm (IIK-2401A/B/C)	① Check the torque, vibration and ampere in the field. ② Adjust an amount of slurry fed to the centrifuge. ③ Check a plugging of lower chute and ML line of the centrifuge with TA. ④ Purge the centrifuge with LPW.

Equipment	Factor	Operational Control Figure			Reason for Setting of Critical Figure	Method of Detection	Example of Action to be Taken upon Deviation of Control Figure from Normal One
		Normal	Allowable	Critical			
(PM-401 A/B/C)							⑤ In case where the ampere is not returned to the normal figure even with the actions taken above, switch the centrifuge over to the stand-by side.
	7) <u>Reslurry Water</u> Lower Limit (FICK-2401A/B/C)	33.7t/hr (per one unit)	31t/hr (Alarm)	30t/hr	Slurry fed to PM-401 is separated into TA solid and ML. TA so separated is reslurried with LPW in PD-401. A shortage of reslurry water may lead to a degradation of product quality and a plugging of the lower chute of PM-401, resulting in stoppage of PM-401 due to overload.	Flow Indicator and Regulator with Alarm (FICK-2401A/B/C)	① Confirm that the indication on FICK-2401A/B/(C) are correct by comparing them with those on LIC-2401, LICK-2402 and FIK-2502. ② Check the working condition of FV-2401 A/B/(C), and adjust the flow rate of reslurry by controlling them on manual. ③ Switch PM-401 over to the stand-by side, if any problem with PM-401A(B).
27. Reslurry Drum (PD-401)	1) <u>Pressure</u> Upper Limit (PICK-2401)	4.2 kg/cm ² G	4.6 kg/cm ² G (Alarm)	7.0 kg/cm ² G (Design)	It is specified to prevent PD-401 from being damaged due to overpressurization.	Pressure Indicator and Regulator with Alarm (PICK-2401)	① Take the same actions as those indicated in the item 23.-1)-① to ③ above.
	2) <u>Liquid Level</u> Upper Limit (LIC-2401)	20%	40% (Alarm)	80%	It is specified to prevent the slurry from overflowing as well as to avoid a possibility of plugging the bottom of PM-401 with TA.	Level Indicator and Regulator with Alarm (LIC-2401)	① Confirm that the indication on LIC-2401 is correct by comparing it with that on LG-2401. ② Check the working condition of LV-2401 and adjust the level by controlling LIC-2401 on manual, if necessary. Also, confirm that the specified LPW is introduced through FIC-2705 to LV-2401.

Equipment	Factor	Operational Control Figure			Reason for Setting of Critical Figure	Method of Detection	Example of Action to be Taken upon Deviation of Control Figure from Normal One
		Normal	Allowable	Critical			
(PD-401)							③ Check the indications on LICK-2403A, LICK-2404A or B and IIK-2401A/B/(C) to be sure that the ML doesn't fall to PD-401 from PM-401. ④ In case where the level is not returned to the normal figure even with the actions taken above, purge the discharge line of PD-401 including LV-2401 with LPW.
28. Agitator (PJ-401)	1) Seal Pressure Lower Limit (PICS-2735)	9 kg/cm ² G	7.0 kg/cm ² G (Alarm)	6.0 kg/cm ² G Interlock	It is specified due to the same reason as mentioned in the item 13. PJ-301.	Pressure Indicator and Regulator with Alarm and Interlock (PICS-2735)	① Upon receiving the alarm sound, check the operating condition of PP-504 and confirm that PIC-2504 and PICS-2735 are functioning properly. ② At the pressure of 6.0 kg/cm²G on PICS-2735, SDV-2735 is closed.
29. Vacuum Filter Feed Drum (PD-402)	1) Liquid Level (1) Upper Limit (LICK-2402A/B)	15%	35% (Alarm)	45% Interlock	It is specified to prevent TA solid from being entrained with vapor over to PD-402 vent system.	Level Indicator and Regulator with Alarm and Interlock (LICK-2402A/B)	(1)-① Confirm that the indication on LICK-2402 is correct by comparing it with that on LG-2402. ② Check the indications on LICK-2402 A/B and XIK-2402A/B, adjust the flow rate of slurry fed to PM-402 by adjusting XIK-2402A/B. ③ Confirm that an excessive amount of slurry is not discharged to PD-402 from PD-401. ④ When the level on LICK-2402 rises to 45%, the slurry discharged through LV-2401 to PD-402 is halted by the actuation of the Interlock.

Equipment	Factor	Operational Control Figure			Reason for Setting of Critical Figure	Method of Detection	Example of Action to be Taken upon Deviation of Control Figure from Normal One
		Normal	Allowable	Critical			
(PD-402)	(2) Lower Limit (LICK-2402A/B)	15%	7% (Alarm)	5% (Alarm)	If the nozzle installed on the overflow line, which serves to return the slurry overflowing the feed pan of PM-402 to PD-402, emerges over the liquid level in PD-402, the vent gas in PD-402 goes back to PM-402 reversely.	Level Indicator and Regulator with Alarm (LICK-2402A/B)	(2)-① Take the same actions as mentioned in the items 24.-(1)-①-② above. ② Confirm that the slurry is smoothly discharged through LV-2401 to PD-402.
	2) Pressure Upper Limit (PIS-2406)	0.04~0.13 kg/cm ² G	0.2 kg/cm ² G (Alarm)	0.25 kg/cm ² G (Interlock)	It is specified to prevent PD-402 from being overpressurized due to a plugging of the vent line.	Pressure Indicator with Alarm (PIS-2406)	① Confirm that the indication on PI-2406 is correct by comparing it with that on PG-2409. ② Confirm that an excessive amount of slurry is not discharged to PD-402 from PD-401. ③ Compare the indication on PG-2409 with that on PIS-2510 for PT-501 and confirm that the vent line of PD-402 to PT-501 is not plugged with TA. ④ Also, check the indication on FG-2755 to be sure that the specified rate of LPW is fed to the vent line of PD-402. ⑤ When the vent pressure on PIS-2406 rises to 0.25 kg/cm ² G, the slurry discharged to PD-402 from PD-401 is halted by the actuation of the Interlock.
30. Mother Liquor Drum (PD-403)	1) Pressure Upper Limit (PIC-2401)	4.2 kg/cm ² G	4.6 kg/cm ² G (Alarm)	7.0 kg/cm ² G (Design)	It is specified to prevent PD-403 from being overpressurized.	Pressure Indicator and Regulator with Alarm (PIC-2401)	① Take the same actions as mentioned in the items 21. Pressure Centrifuge-1) pressure.

Equipment	Factor	Operational Control Figure			Reason for Setting of Critical Figure	Method of Detection	Example of Action to be Taken upon Deviation of Control Figure from Normal One
		Normal	Allowable	Critical			
(PD-403)	2) <u>Liquid Level</u> (1) Upper Limit (LICK-2403A/B)	30%	40% (Alarm)	80%	(1) It is specified to prevent a liquid in PD-403 from overflowing as well as to prevent PM-401 from becoming impossible of discharging ML to PD-403.	Level Indicator and Regulator with Alarm (LICK-2403A/B)	(1)-① Check the indications on II-2401A/B (C), XJI-2401A/B(C) and LICK-2304, and confirm that an excessive amount of slurry is not fed to PM-401. ② Check the working condition of LV-2403A to be sure that the discharge line from PD-403 to PD-404 is not plugged with solid crystallized. ③ If the level on LIC-2403A is not returned to the normal condition even with the action above, blow a liquid in PD-403 to the Blow-down Pit through LV-2403B and wash LV-2403A with LPW and caustic.
	(2) Lower Limit (LICK-2403A/B)	30%	20% (Alarm)	5%	It is specified to prevent the gas phase in PD-403 from being discharged directly through LCV-2403A to PD-404.	Ditto	(2)-① Confirm that an excessive amount of liquid is not discharged through LCV-2403A to PD-404 from PD-403. And, adjust the level of PD-403 by controlling LICK-2403A on manual, if necessary. ② Confirm that the indication on LICK-2403A is correct by comparing it with that on LG-2403.

Equipment	Factor	Operational Control Figure			Reason for Setting of Critical Figure	Method of Detection	Example of Action to be Taken upon Deviation of Control Figure from Normal One
		Normal	Allowable	Critical			
31. Agitator (PJ-402)	<u>Ampere</u> Upper Limit (Ampere Meter in the field)	23A	30A (Rated)	30A (Rated)	It is specified to prevent the motor from being damaged due to the overload.	Ampere Meter with Stop Alarm	① Check the indication on LICK-2402. ② Check to be sure that there is not any problem with the motor and reducer. ③ If it is judged impossible to continue operation based on the facts found in the actions above, stop feeding CTA to PD-101 and switch the operation over to the Recycle Solvent Run.
32. ML Flash Drum (PD-404)	<u>Liquid Level</u> Upper Limit (LICK-2404A/B)	30%	60% (Alarm)	80%	It is specified to prevent PE-404 from being fouled with the ML entrained with vapor.	Level Indicator and Regulator with Alarm (LICK-2404A/B)	① Confirm that an excessive amount of liquid is not discharged through LCV-2403A to PD-404 from PD-403. ② Check to be sure that there is not any place clogged with solid on the suction and delivery lines of PP-404 A/B. ③ Check the working condition of LV-2404A(B) and adjust the level by controlling LICK-2404A(B) on manual, if necessary. ④ Switch PP-404 over to the stand-by side.
33. Agitator (PJ-404)	<u>Ampere</u> Upper Limit (IJI-2424)	6~10A	17.9A (Alarm)	17.9A (Rated)	It is specified to prevent the motor from being damaged due to the overload.	Ampere Meter with Alarm (IJI-2424)	① Check the indication on IJI-2424. ② Check the revolution sound of PJ-404 to see if a rise in ampere on IJI-2424 is due to a fall of a lump of solid built on the wall of drum. ③ Check to be sure that there is not any problem with the motor and reducer.

Equipment	Factor	Operational Control Figure			Reason for Setting of Critical Figure	Method of Detection	Example of Action to be Taken upon Deviation of Control Figure from Normal One
		Normal	Allowable	Critical			
(PJ-404)							④ If it is judged impossible to continue operation based on the facts found in the actions above, stop PJ-404. ⑤ Discharge the ML in PD-403 through LV-2403B to the Blow-down Pit TU-802, during which time PD-404 is to be washed with caustic and PJ-404 is then repaired.
34.Funda Filter (PM-901)	1) <u>Pressure</u> Upper Limit (PIS-2901)	0~3 kg/cm ² G	3.0 kg/cm ² G	3.0 kg/cm ² G	It is specified to prevent the filter leaves from being damaged due to the overpres-surization.	Pressure Indicator with Alarm and Sequence (PIS-2901)	① When the pressure indicated on PIS-2901 rises to 3.0 kg/cm ² G, the waste water fed to PM-901 is stopped by the Sequence Program. ② Wash PM-901 with caustic by starting the sequence concerned.
	2) <u>Ampere</u> Upper Limit (Ampere Meter in the field)		65A (Rated)	65A (Rated)	It is specified to prevent the motor from being damaged due to the overload.	Ampere Meter with Stop Alarm	① Stop the sequence program to discontinue the operation of PM-901 and put it into caustic washing. ② Check the motor to see if there is no problem with the motor. ③ If PM-901 can not be turned by hand, disassemble PM-901 to check for any defect.

Equipment	Factor	Operational Control Figure			Reason for Setting of Critical Figure	Method of Detection	Example of Action to be Taken upon Deviation of Control Figure from Normal One
		Normal	Allowable	Critical			
35.Rotary Vacuum Filter (PM-402 A/B)	1) <u>Liquid Level</u> Upper Limit (LIS-2400A/B)	60%	a) 80% (Alarm)	b) 95% Interlock	If the overflow line to recycle a portion of slurry through PM-402 to PD-402 is plugged, slurry enters the dryer overflowing the weir in the feed pan of PM-402.	Level Indicator with Alarm and Interlock (LIS-2400A/B)	a)-① Check the flow rate of slurry fed to PM-402 indicated on FICK-2402 A/B. ② Check the overflow line for its plugging and empty in PM-402A/B and then replace it with LPW. b)-① When the level on LIS-2400A/B rises to 95%, the slurry feed (FV-2402A/B), LPW rinse (KCV-2426A/B) and filter flashing sequence (SC-2402) are stopped by the actuation of the Interlock. ② In case where the plugging of overflow line can not be solved even with the action taken in the item a)-② above, wash it with caustic while accumulating slurry in PD-402.
	2) <u>Ampere</u> Upper Limit (IJIK-2411A/B)	7A	13.2A (Alarm)	13.2 (Rated)	It is specified to prevent the motor from being damaged due to the overload.	Ampere Meter with Alarm (IJIK-2411A/B)	① Check the feed rate on FICK-2402A/B and revolution speed on XIK-2402 A/B, etc. ② Check the motor and the body of PM-402 for any abnormalities such as vibration and abnormal sound, etc. ③ Stop PM-402 to repair in case of its being different to continue operation.

Equipment	Factor	Operational Control Figure			Reason for Setting of Critical Figure	Method of Detection	Example of Action to be Taken upon Deviation of Control Figure from Normal One
		Normal	Allowable	Critical			
(PM-402 A/B)	3) <u>Vacuum</u> Upper Limit (PICK-2405)	-200~ -450 mmHg	-500mmHg (Alarm)	-600mmHg	It is specified to monitor the extent of plugging of the filter cloth.	Pressure Indicator and Regulator with Alarm (PICK-2405)	① Confirm that the indication on PICK-2405 is correct by comparing it with that on PG-2415. ② When the indication on PICK-2405 reaches the specified figure, stop PM-402 and wash it with caustic.
	4) <u>Blow-Back Pressure</u> (1) Upper Limit (PICK-2402A/B)	0.12 kg/cm ² G	0.17 kg/cm ² G (Alarm)	0.2 kg/cm ² G	It is specified to prevent a damage of filter cloth due to a high pressure.	Pressure Indicator and Regulator with Alarm (PICK-2402A/B)	(1)-① Check the working condition of PV-2402A(B) and adjust the blow-back pressure by controlling PICK-2402A(B) on manual, if necessary. ② Check the working condition of PC-403 for its abnormality. ③ If there is any problem with PICK-2402A(B), stop PM-402 and repair it.
	(2) Lower Limit (PICK-2402A/B)	0.12 kg/cm ² G	0.11 kg/cm ² G (Alarm)	0.09 kg/cm ² G	It is specified to prevent a TA cake on filter cloth from being left over due to a low blow-back pressure.	Ditto	(2)-① Take the same action as mentioned in the item (1)-① above.
36. Filtrate Separating Drum (PD-405)	<u>Liquid Level</u> Upper Limit (LICK-2405)	15%	(1) 25% (Alarm)	(2) 40% Interlock	When the liquid level of PD-405 rises, the filtrate discharged from PM-402 is to be obstructed with a result that it leads to a decrease in an amount of gas sucked in PC-403A, and the overload of PC-403A due to a liquid entrained with gas in the worst case.	Level indicator and Regulator with Alarm (LICS-2405)	(1)-① Confirm that the indication on PICS-2405 is correct by comparing it with that on PG-2415. ② Check LV-2405 for its function and adjust the level by controlling LICS-2405 on manual.

Equipment	Factor	Operational Control Figure			Reason for Setting of Critical Figure	Method of Detection	Example of Action to be Taken upon Deviation of Control Figure from Normal One
		Normal	Allowable	Critical			
(PD-405)							③ Check to see if there is any plugging on the suction of PP-405 A/B. And if any, change the suction of it from the bottom to the side suction. (2)-① When the level on LICS-2405 reaches 40%, PV-2405 is opened and vacuum on PM-402 is cut by the actuation of the Interlock. ② If the plugging can not be solved with the LPW purge, stop feeding slurry to PM-402 and wash it with caustic.
37.Vacuum Pump (PC-403)	1) <u>Ampere</u> Upper Limit (IJI-2403)	95~105A	118A (Rated)	118A (Rated)	It is specified to prevent the motor from being damaged due to the overload.	Ampere Indicator with Alarm (IJI-2403)	① Check the flow rate of sealant to PC-403A and adjust it by controlling FIC-2403 concerned. ② Check PC-403A for abnormal sound, vibration, tension of belt and temperature rise of motor, etc. ③ If the said PC-403A is judged to have any problem, change it to the stand-by.
	2) <u>Sealant Flow</u> (1) Upper Limit (FIC-2403) (2) Lower Limit (FIC-2403)	21t/hr 21t/hr	22t/hr (Alarm) 19t/hr (Alarm)	23t/hr	If an excess amount of sealant is fed to PC-403A, there is a case to cause such troubles as an abnormal sound and vibration, etc., resulting in overload. It is specified as the minimum flow required for removing the compression heat which is generated by PC-403A in operation.	Flow Indicator and Regulator with Alarm (FIC-2403) Ditto	(1)-① Check FV-2403 for its working condition and adjust the flow rate by controlling it on manual. ② Check the pressure indicated on PG-2412. (2)-① Take the same actions as mentioned in the item (1) above.

Equipment	Factor	Operational Control Figure			Reason for Setting of Critical Figure	Method of Detection	Example of Action to be Taken upon Deviation of Control Figure from Normal One
		Normal	Allowable	Critical			
38.Vacuum Pump Separating Drum (PD-406)	<u>Liquid Level</u> Upper Limit (LIC-2406)	20%	30% (Alarm)	40%	If the liquid level of PD-406 rises, the outlet line of PC-403A is obstructed with a result that it leads to a decrease in the gas discharged from PC-403A and the overload of PC-403A in the worst case. (Note)The position just below the gas inlet nozzle on PD-406 is equivalent to the level of approx. 40% indicated on LIC-2406.	Level Indicator and Regulator with Alarm (LIC-2406)	① Check the indication on LIC-2406 for its correctness by comparing it with that on LG-2406. ② Check the working condition of LV-2406, and adjust the liquid level by controlling LIC-2406 on manual. ③ Check the flow rate of sealant fed to PC-403A on FIC-2403. ④ Blow down the liquid in PD-406 through the by-pass line of LV-2406, if necessary.
39.Dryer (PM-404)	1) <u>Outlet Temperature</u> Lower Limit (TI-2410)	132°C	125°C (Alarm)	120°C	A fall of temperature on the outlet of dryer leads to an increase in the volatile content in the PTA product.	Temperature Indicator with Alarm (TI-2410)	① Increase the set pressure of steam fed to the dryer through PIC-2404A. ② Check the operating condition of PM-402; a) Check the indications on FICK-2402 and LICK-2402 to be sure that the specified amount of slurry is being treated by PM-402. b) Check PM-402 to see if there is no slurry overflowing the weir to the dryer. c) Check the wetness of wet cake fed to the dryer based on the indications on FI-2407 and FI-2409. ③ Check the operating conditions of the crystallizer train to see if there is no possibility of causing the APS of PTA product to vary due probably to a variation of operating conditions such as the temperature and level.

Equipment	Factor	Operational Control Figure			Reason for Setting of Critical Figure	Method of Detection	Example of Action to be Taken upon Deviation of Control Figure from Normal One
		Normal	Allowable	Critical			
(PM-404)							④ Confirm the steam trap (ST-2415) is functioning properly. ⑤ Check the carrier N₂ gas flow indicated on FICS-2411 and its temperature on TJI-2414. ⑥ Change the heating steam from 3S(PV-2404A) to PZ-409(PV-2404B). ⑦ In case where the heating tube of dryer has been fouled; a) Stop feeding slurry to RVF, operate the dryer under no load condition, paying attention to the temperature on TI-2410. b) If the performance of dryer has not been restored to the normal condition in spite of no load operation above, wash the dryer with caustic.
	2) <u>Steam Flow</u> Upper Limit (FI-2404)	5-6t/hr	6.5t/hr (Alarm)	7.0t/hr	An increase in an amount of heating steam means an increase of wetness contained in wet cake and an introduction of slurry overflowing the weir to the dryer. Accordingly, it is specified to promptly detect the abnormal conditions and to prevent a deterioration of product quality.	Flow Indicator with Alarm (FI-2404)	① Check PM-402 to see if there is no liquid overflowing the weir to the dryer. ② Check the wetness of wet cake fed to the dryer based on the indications on FI-2407 and FI-2409. ③ Confirm that an appropriate amount of slurry is being treated by PM-402. And adjust the said feed rate to PM-402 by controlling FICK-2402A(B), if necessary. ④ Confirm that the indication on FI-2404 is correct and that the steam trap is working properly. ⑤ Check the liquid level of PT-401 to see if there is any overflow to the dryer.

Equipment	Factor	Operational Control Figure			Reason for Setting of Critical Figure	Method of Detection	Example of Action to be Taken upon Deviation of Control Figure from Normal One
		Normal	Allowable	Critical			
(PM-404)							⑥ Check the operating conditions for the dryer system whether there is any damage on the heating tube or not. If any damage on the heating tube, move to the ESD.
	3) <u>Dryer Pressure</u> Upper Limit (PI-2412)	600mmAq	700mmAq (Alarm)	800mmAq (Seal Pot)	It is specified to prevent water vapor and TA from spouting from the seal system of dryer owing to an excess pressure. (Note) Dryer designed by the pressure of 0.1 kg/cm ² G is protected by the seal pot having the seal leg of 800mmAq in depth.	Pressure Indicator and Regulator with Alarm (PI-2412)	① Confirm that there is no plugging for the line connecting PM-403 and PT-401, and the trays in PT-401 by comparing the indication on PI-2412 with that on PI-2411. ② Check the indication on FIC-2411 and adjust rate of carrier gas fed to dryer, if necessary. ③ Take the same actions as explained in the item 2) Steam Flow above.
	4) <u>Carrier Gas Flow</u> Lower Limit (FICS-2411)	1,500 Nm ³ /hr	1,000 Nm ³ /hr (Alarm)	250Nm ³ /hr Interlock	A shortage of amount of carrier gas leads to an inferior drying of product.	Flow Indicator with Alarm (FICS-2411)	① Check the indication on FIC-2411. ② Check the working condition of FV-2411 and adjust the flow rate of carrier gas by controlling FIC-2411 on manual. ③ Check the N ₂ pressure on PG-3127. ④ When the indication on FIC-2411 falls down to 250Nm ³ /hr, the dryer system gets into the shutdown, thereby stopping feeding slurry to PM-402.
	5) <u>WG Flow</u> Lower Limit (FIC-2405)	5,295 Nm ³ /hr	5,000 Nm ³ /hr (Alarm)	4,800 Nm ³ /hr Interlock	It is specified to prevent the plugging in the pneumatic transportation line due to a shortage of the WG flow.	Flow Indicator and Regulator with Alarm and Indicator (FICS-2405)	① Check the WG flow rate on FICS-2405 and 7N pressure on PI-2800. ② Check the working condition of FV-2405 and adjust the WG flow rate by controlling FICS-2405 on manual.

Equipment	Factor	Operational Control Figure			Reason for Setting of Critical Figure	Method of Detection	Example of Action to be Taken upon Deviation of Control Figure from Normal One
		Normal	Allowable	Critical			
(PM-404)							③ By-passing the Interlock for the dryer system, shut KCV-2450, stop PZ-405 and PZ-403 temporarily and purge the pneumatic transportation line by raising the flow rate on FICS-2405 a little. (Note: D.P. of pipe line; 4.0 kg/cm²G) ④ When the indication on FICS-2405 falls down to 4,800Nm³/hr, the dryer system gets into the SD condition by the actuation of the Interlock. ⑤ In case where the items mentioned in the items ③ and ④ above requires a long time, switch the operation over to the Recycle Slurry Operation.
	6) <u>O₂ Content</u> Upper Limit (AI-2400)	4.0vol%	7.0vol% (Alarm)	8.0vol%	TA powder in the gas containing high concentration of O ₂ has a possibility of causing dust explosion.	O ₂ Content Indicator with Alarm (AI-2400)	① Check AI-2400 for its zero point, span and the flow rate of sample gas. ② In case where PC-402 is being operated: a) Check the flow rate of N₂ gas fed to the labyrinth seal of PC-402. b) Check to be sure that there is no air sucked in the suction of PC-402, which is operated under negative pressure. c) Check the pressure control on the silo system. d) Check the differential pressure on the bag filter.

Equipment	Factor	Operational Control Figure			Reason for Setting of Critical Figure	Method of Detection	Example of Action to be Taken upon Deviation of Control Figure from Normal One
		Normal	Allowable	Critical			
40. Rotating Equipment (PM-404-2)	<u>Ampere</u> Upper Limit (IJI-2404B)	6.5~7.5A	9.0A (Alarm)	13.2A (Rated)	It is specified to prevent the motor from being damaged due to the overload. And, further, the Interlock is provided to prevent the dryer system from being plugged with TA upon its stoppage.	Ampere Indicator with Alarm (IJI-2404B)	① Confirm that there is no plugging at the inlet of PM-404. ② Check to see if there is not a lot of TA discharged from PM-402. ③ Stop feeding slurry to PM-402 temporarily. ④ Check the working condition for the motor and reducer, and if it is judged to be impossible to continue operation, stop feeding CTA to PD-101 and switch the operation over to the Recycle Slurry Operation. ⑤ In case where PM-404-2 stopped due to the overload, the slurry feed to PM-402 is stopped by the actuation of the Interlock. And so, if it is expected to take a long time to restart up PM-404-2, switch the operation over to the Recycle Slurry Operation.
41. Rotating Equipment (PM-404-1)	<u>Ampere</u> (a) Upper Limit (IJI-2404A)	70~90A	100A (Alarm)	126A (Rated)	Ditto	Ampere Indicator with Alarm (IJI-2404A)	(a) & (b) ① Take the same actions as mentioned in the items ②, ③, ④ & ⑤ above.
(PZ-403)	b) Upper Limit (Ampere Meter)	1.5A	6.5A (Rated)	6.5A (Rated)	Ditto		② Check the indications on PIK-2403 and the ampere of PZ-403 to be sure that an appropriate amount of PTA product is being discharged from the dryer. Further, check the indication on FI-2404 to see if there is not a lot of product accumulated in the dryer.

Equipment	Factor	Operational Control Figure			Reason for Setting of Critical Figure	Method of Detection	Example of Action to be Taken upon Deviation of Control Figure from Normal One
		Normal	Allowable	Critical			
(PZ-403)							(Note) Symptoms on the accumulation of product in the dryer; a) Decrease in the pressure on PIK-2403. b) Decrease in the ampere of PZ-403. c) Increase in an amount of steam on FI-2404. d) Increase in the temperature on TI-2410. ③ If both PM-404-1 and PZ-403 stop due to the overload, slurry fed to PM-402 is stopped by the actuation of the Interlock. In case where the restoration of PM-404-1 (PZ-403) is expected to be delayed for a long time, proceed to the Recycle Slurry Operation by stopping feeding CTA to PD-101. ④ In the event of shear pin cut for PZ-403, the same action as mentioned above is to be taken.
42. Blower (PC-402)	<u>Ampere</u> Upper Limit (Ampere Meter)	36A	48A (Rated)	48A (Rated)	It is specified to prevent the motor from being damaged due to the overload.	Ampere Meter with Stop Alarm	① Check the flow rate of carrier gas on FICS-2405 and adjust it, if necessary. ② Check the operating condition for the motor.
43. Dryer N ₂ Scrubber (PT-401)	1) <u>Liquid Level</u> Upper Limit (LICS-2410)	50%	60% (Alarm)	80% Interlock	It is specified to prevent a liquid in PT-401 from overflowing through PM-403 to the dryer.	Level Indicator and Regulator with Alarm (LICS-2410)	① Confirm that the indication on LICS-2410 is correct by comparing it with that on LG-2411. ② When the indication on LICS-2410 reaches 80%, FV-2408 is closed. Then, its controller is returned to the normal control at the level of 50% on LICS-2410.

Equipment	Factor	Operational Control Figure			Reason for Setting of Critical Figure	Method of Detection	Example of Action to be Taken upon Deviation of Control Figure from Normal One
		Normal	Allowable	Critical			
(PT-401)							③ Check the working condition of LV-2410, and purge it with LPW if there is any plugging with it. ④ If LV-2410A was impossible to operate, change to LV-2410B from its CV, and then repair LV-2410A.
	2) <u>Pressure</u> (Carrier Gas) Upper Limit (PI-2411)	600mmAq	690mmAq (Alarm)	800mmAq (Seal pot)	It is specified to promptly detect the plugging on PT-401 as well as to prevent the equipment from damaging due to an excess pressure.	Pressure Indicator with Alarm (PI-2411)	① Confirm that the indication on PI-2411 is correct by comparing it with that on PI-2412. ② Check to be sure that an excess amount of N ₂ is not fed to the dryer system by checking the indications on FIC-2411, FG-2405 and FG-2406. ③ Check to be sure that there is no plugging with PT-401 tray. ④ Take the same actions as mentioned in the item 35.②-⑤ above.
	3) <u>Sprayed DIW</u> Lower Limit (FIC-2408)	14t/hr	11t/hr (Alarm)	10t/hr	A decrease in an amount of DIW sprayed at the top of PT-401 worsens the efficiency of collecting fine particle entrained with carrier gas. Also, the said above leads to one of causes to plug the tray in PT-401.	Flow Indicator and Regulator with Alarm (FIC-2408)	① Confirm that the indication on FIC-2408 is correct by comparing it with those on FI-2409 and FI-2404. ② Check the working condition of FV-2408 and adjust the flow rate of DIW fed to PT-401 by controlling FIC-2408 on manual. ③ If there is any problem with FV-2408, operate PT-401 bypassing it and repair.

Equipment	Factor	Operational Control Figure			Reason for Setting of Critical Figure	Method of Detection	Example of Action to be Taken upon Deviation of Control Figure from Normal One
		Normal	Allowable	Critical			
44.PTA Product Silo (PTK-800 A/B/C)	1) <u>Pressure</u> (1)Upper Limit PICK-2801 A/B/C	70~500 mmAq	600mmAq (Alarm)	800mmAq (PRV-2806 A/B/C)	(1)They are specified to prevent the silo system from damaging due to overpres-surization.	Pressure Indicator and Regulator with Alarm (PIC-2801A/B/C)	(1)- ① Confirm that PIC-2801A(B)(C) is functioning properly. ② Check the working condition of PV-2801A/B/C. ③ Check to be sure that there is no plugging with the bag filter. ④ Switch the silo over to the stand-by side to reconfirm an abnormality.
	(2)Lower Limit (PICS-2802)	200mmAq	0mmAq (Alarm)	-40mmAq Interlock	(2)It is specified to prevent the silo from damaging due to an excessive vacuum while the blower is being run.	Pressure Indicator And Regulator with Alarm and Interlock (PICS-2802)	(2)- ① Check the differential pressure on the bag filter PM-803A/B/C. ② Check the flow rate of waste gas for use in pneumatic transportation and adjust it, if necessary. ③ Check the working condition of the pressure control valves PV-2801A/B/C and PV-2802. ④ When the indication on PICS-2802 falls down to -40mmH ₂ O, the blower PC-402 is stopped by the actuation of the Interlock.
45.Bag Filter (PM-803 A/B/C)	<u>Differential Pressure</u> Upper Limit (PDI-2803 A/B/C)	30mmAq	150mmAq (Alarm)	250mmAq	The filter cloth and filter retainer may be collapsed by the differential pressure of 250 mmAq or higher.	Differential Pressure Indicator with Alarm (PDI-2803A/B/C)	① Check to be sure that an excess amount of WG is not used for pneumatic transportation. ② Check the working condition of the pulse unit and the pressure of N ₂ fed to the unit. ③ Replace the filter cloth with the new one, if it is considered the end of life.

Equipment	Factor	Operational Control Figure			Reason for Setting of Critical Figure	Method of Detection	Example of Action to be Taken upon Deviation of Control Figure from Normal One
		Normal	Allowable	Critical			
46.Vent Scrubber (PT-501)	1) <u>Pressure</u> (1) Upper Limit (PIS-2510)	200~280 mmAq	a) 1,000 mmAq Interlock	b) 3,000 mmAq (Seal Pot)	PT-501 serves as the vent scrubber for the crystallizer train and also as the receiver for the vapor relieved from the safety valves installed on the reactor and crystallizers. And so, it is specified to prevent the vent scrubber from being damaged due to the overpressurization.	Pressure Indicator with Alarm And Interlock (PIS-2510)	(1)-a) ① When the indication on PIS-2510 rises to 1,000mmAq, the by-pass valve of PE-501 (SDV-2510B) is opened by the actuation of the Interlock. ② Check the opening degrees for the pressure control valves relating to the crystallizer train. ③ Check to see if there is any clogging with PT-501 tray and PE-501 tube. (1)-b) ① When the indication on PIS-2510 rises to 3,000mmAq or more, the seal pot PZ-501 spouts to release the pressure of PT-501. ② Confirm that any of the safety valves installed on the reactor and crystallizers is not relieving pressure. (SV-2201, SV-2301-2304) ③ Move to the ESD depending on the situation.
	(2) Lower Limit (PIS-2510)	200~280 mmAq	a) 130mmAq (Alarm)	b) 100mmAq Interlock	An excess amount of H ₂ is being discharged through PZ-502 to the atmosphere after being diluted with the steam. Consequently, it is specified to have a possibility that H ₂ gas forms an explosive gas mixed with air sucked in PT-501 due to an excessive low pressure.	Pressure Indicator with Alarm And Interlock (PIS-2510)	(2)-a) ① Check the opening degrees for the pressure control valves relating to the crystallizer train. ② Decrease an amount of heat recovered by the crystallizer train. (2)-b) ① By the actuation of the Interlock, 3S is introduced into the inlet of PE-501 through SDV-2510A.

Equipment	Factor	Operational Control Figure			Reason for Setting of Critical Figure	Method of Detection	Example of Action to be Taken upon Deviation of Control Figure from Normal One
		Normal	Allowable	Critical			
47.Recycle Solvent Drum (PD-501)	1)Liquid Level (1)Upper Limit (LIC-2501A/B)	60%	a) 70% (Alarm)	b) 80% Interlock	It is specified to prevent the hot water from spouting to the atmosphere due to an excessive rise of level.	Level Indicator and Regulator with Alarm (LIC-2510A/B)	(1)-a) ① Adjust the liquid level by controlling LIC-2501A & B on manual. ② Check the working condition of LV-2501A/B. ③ Check an amount of solvent sent from PD-405 and PT-401 (FI-2407 and FI-2409) via PE-403. ④ Check an amount of solvent transferred to PD-101 from PD-501. (FICK-2102) ⑤ Adjust an amount of solvent recovered from PT-501 to PD-501.
	(2)Lower Limit (LIC-2501A/B)	60%	50% (Alarm)	40%	It is specified to prevent PP-501 from getting into the cavitation phenomenon as well as to avoid leading to the shutdown due to a shortage of the recycle solvent.	Level Indicator and Regulator with Alarm (LIC-2510A/B)	(1)-b) ① By the actuation of the Interlock, LV-2501A is closed and LIC-2501B, MO=0 and to automatic from manual. (2)-① Take the same actions as mentioned in the items (1)-a)-①-⑤ above. ② In case where the recycle solvent can not be secured, stop feeding CTA to PD-101 and switch the operation over to the LPW run.
	2)Temperature Lower Limit (TJI-2501)	100°C	95°C (Alarm)	94°C	As a temperature of solvent in PD-501 falls, O ₂ content dissolved in solvent increases, thereby being one of causes generating Stress Corrosion Cracking. (S.C.C.) Note; O ₂ Content Dissolved in DIW; 20°C 9 ppm 100°C 0 ppm	Temperature Indication with Alarm (TJI-2501)	① Check the temperature of the recycle solvent heated through PE-403, which is indicated on TIC-2403. ② Check the temperature of LPW made up for the process water, which is indicated on TIC-2502 and TI-2507.

Equipment	Factor	Operational Control Figure			Reason for Setting of Critical Figure	Method of Detection	Example of Action to be Taken upon Deviation of Control Figure from Normal One
		Normal	Allowable	Critical			
48.Process Water Drum (PD-502)	1) <u>Liquid Level</u>						
	(1) Upper Limit (LIC-2502)	80%	90% (Alarm)	95%	It is specified to prevent the hot water from spouting to the atmosphere due to an excessive rise of level.	Level Indicator and Regulator with Alarm (LIC-2502)	1)-① Adjust the liquid level by controlling LIC-2502 on manual. ② Check the working condition of LV-2502. ③ Check an amount of DIW received from the B/L. (FIQK-2500)
	(2) Lower Limit (LIC-2502)	80%	70% (Alarm)	50%	It is specified to prevent pump from getting into the cavitation as well as to avoid leading to ESD due probably to halting a flushing to the seal system.	Level Indicator and Regulator with Alarm (LIC-2502)	(2)-① Take the same actions as mentioned in the items (1)-① to ③ above. ② Check the indication on PDI-2500 to be sure that the DIW FILTER PM-500 is not plugged. ③ Confirm that the DIW pressure on PG-3303 is the normal one. ④ In case where DIW can not be received any more due to a trouble in the Utility plant, move to the plant ESD.
	2) <u>Temperature</u>						
	Lower Limit (TI-2507)	100°C	95°C (Alarm)	94°C	As a temperature of solvent in PD-502 falls, O ₂ content dissolved in solvent increases, thereby being one of causes of generating Stress Corrosion Cracking.	Temperature Indicator with Alarm (TI-2507)	① Check TIC-2502 and the steam trap (ST-2502) to be sure that they are functioning properly. ② Check the indication on TJI-2506 installed on the outlet of PE-404 to be sure that the heat recovery is being carried out normally via PE-501 to PE-404. ③ Check to see if an excess amount of process water is not used for any user.

Equipment	Factor	Operational Control Figure			Reason for Setting of Critical Figure	Method of Detection	Example of Action to be Taken upon Deviation of Control Figure from Normal One
		Normal	Allowable	Critical			
49. DIW Filter (PM-500 A/B)	<u>Differential pressure</u> Upper Limit (PDI-2500)	1.45 kg/cm ² G>	1.5 kg/cm ² G (Alarm)	1.6 kg/cm ² G	It is specified to prevent the filter element from being damaged due to a high differential pressure.	Differential Pressure Indicator with Alarm (PDI-2500)	① Switch the filter over to the stand-by side and clean it.
50. Process Water Heater (PE-502)	<u>Temperature</u> (1) Upper Limit (TIC-2502)	101.5°C	a) 108°C (Alarm)	b) 130°C (Design)	It is specified to prevent the hot water from spouting to the atmosphere due to an excessive temperature rise. Also, the tube side for PE-502 is designed by the temperature of 130°C	Temperature Indicator and Regulator with Alarm (TIC-2502)	(1)-a)-① Adjust the temperature at the outlet of PE-502 by controlling TIC-2502 on manual. ② Check the working condition of TV-2502. ③ Check the pressure of steam fed to PE-502. ④ Check the variation of usage of the process water.
	(2) Lower Limit (TIC-2502)	101.5°C	95°C (Alarm)	94°C	As a temperature of solvent in PD-502 falls, O ₂ content dissolved in solvent increases, thereby being one of causes generating S.C.C.	Ditto	(1)-b)-① Stop feeding steam to PE-502 at the temperature of 130°C or higher. (2)-① Take the same actions as mentioned in the items (1)-a)-① ~ ④ above. ② Check the working condition of the steam trap. (ST-2502)
51. MPW Heater (PE-503)	1) <u>Temperature</u> (1) Upper Limit (TIC-2503)	200°C	a) 215°C (Alarm)	b) 220°C (Design)	(1) It is specified to prevent the tube side of PE-503 from being heated over the design temperature.	Temperature Indicator and Regulator with Alarm (TIC-2503)	(1)-a) ① Adjust the temperature at the outlet of PE-503 by controlling TIC-2503 on manual. ② Check the working condition of TV-2503. ③ Check the pressure of steam fed to PE-503. ④ Check the variation of usage of MPW indicated on FIK-2503.

Equipment	Factor	Operational Control Figure			Reason for Setting of Critical Figure	Method of Detection	Example of Action to be Taken upon Deviation of Control Figure from Normal One
		Normal	Allowable	Critical			
(PE-503)							(1)-b) ① Stop feeding steam to PE-503 at the temperature of 220°C or higher.
	(2) Lower Limit (TIC-2503)	200°C	190°C (Alarm)	185°C	(2) As the temperature of MPW falls, the effect of MPW purge worsens that much, thereby resulting in frequent clogging of the pipe line to be used for slurry transportation.	Temperature Indicator and Regulator with Alarm (TIC-2503)	(2)-① Take the same actions as mentioned in the items (1)-a)-① ~ ④ above. ② Check the working condition of the steam trap. (ST-2503)
	2) <u>Pressure</u>						
	(1) Upper Limit (PIC-2503)	53 kg/cm ² G	56 kg/cm ² G (Alarm)	56.5 kg/cm ² G (Design)	(1) It is specified to prevent the tube side of PE-503 from being pressurized over the design pressure.	Pressure Indicator and Regulator with Alarm (PIC-2503)	(1) Adjust the pressure of MPW by controlling PIC-2503 on manual. When the pressure of MPW exceeds 56.5 kg/cm ² G, SV-2503 actuates to relieve the pressure.
	(2) Lower Limit (PIC-2503)	53 kg/cm ² G	46 kg/cm ² G (Alarm)	45 kg/cm ² G	(2) A fall of MPW pressure leads to a decrease in the purging capacity.	Ditto	(2)-① Adjust the pressure of MPW by controlling PIC-2503 on manual. ② Check the working condition of PV-2503. ③ Check the variation of usage of MPW indicated on FICK-2503. ④ Check to be sure that PP-503 is pumping properly. ⑤ Check PIC-2502 to be sure that the pressure of LPW is controlled at the specified one.

Equipment	Factor	Operational Control Figure			Reason for Setting of Critical Figure	Method of Detection	Example of Action to be Taken upon Deviation of Control Figure from Normal One
		Normal	Allowable	Critical			
52. MPW Heater (PE-504)	1) <u>Temperature</u>						
	(1) Upper Limit (TIC-2504)	170°C	a) 180°C (Alarm)	b) 190°C (Design)	(1) If MPW is heated over the temperature on the process side, there is a possibility that instrument malfunctions due possibly to a flash phenomenon to be caused in the conduit pipe.	Temperature Indicator and Regulator with Alarm (TIC-2504)	(1)-a) ① Adjust the temperature by controlling TIC-2504 on manual. ② Check the working condition of TV-2504. ③ Check the pressure of steam fed to PE-504. ④ Check the variation of usage of MPW. (1)-b)-① Since 10K steam is used as the heating medium for PE-504, it may not be heated over 190°C normally.
	(2) Lower Limit (TIC-2504)	170°C	162°C	160°C	(2) As the temperature of MPW falls, the effect of MPW purge worsens that much, thereby resulting in frequent clogging of the pipe line to be used for slurry transportation.	Ditto	(2)-① Take the same actions as mentioned in the items (1)-a)-① ~ ④.
	2) <u>Pressure</u>						
	(1) Upper Limit (PIC-2503)	53 kg/cm ² G	56 kg/cm ² G (Alarm)	56.5 kg/cm ² G (Design)	(1) It is specified to prevent the tube side of PE-504 from being pressurized over the design pressure.	Pressure Indicator and Regulator with Alarm (PIC-2503)	(1) Take the same actions as mentioned in the items 46.-2)-(1) above.
	(2) Lower Limit (PIC-2503)	53 kg/cm ² G	46 kg/cm ² G (Alarm)	45 kg/cm ² G	(2) A fall of MPW pressure leads to a decrease in the purging capacity.	Ditto	(2) Take the same actions as mentioned in the items 46.-2)-(2)-① ~ ⑤ above.

Equipment	Factor	Operational Control Figure			Reason for Setting of Critical Figure	Method of Detection	Example of Action to be Taken upon Deviation of Control Figure from Normal One
		Normal	Allowable	Critical			
53. MPW Heater Steam Drum (PD-1206)	1) <u>Pressure</u> (1) Upper Limit						
	a) 53K MPW System (PIC-2503)	53 kg/cm ² G	56 kg/cm ² G (Alarm)	56.5 kg/cm ² G (Design)	a), b) They are specified to prevent the tube side of PE-504 from being pressurized over the designed pressure.	a) Pressure Indicator and Regulator with Alarm (PIC-2503)	a) Adjust the pressure of MPW by controlling PIC-2503 on manual. When the pressure of MPW exceeds 56.5 kg/cm ² G, SV-2503 actuates to relieve the pressure.
	b) 30K MPW System (PIC-2701)	30 kg/cm ² G	32 kg/cm ² G (Alarm)	33 kg/cm ² G (Design)		b) Pressure Indicator and Regulator with Alarm (PIC-2701)	b)-① Adjust the pressure of MPW by controlling PIC-2701 on manual. ② Check the working condition of PV-2701. ③ Check the pressure on PIC-2503. ④ When the safety valve SV-2607 is actuated, decrease the pressure of 30K MPW system by closing the block valve of PCV-2701 a little.
	(2) Lower Limit a) PIC-2503	53 kg/cm ² G	46 kg/cm ² G (Alarm)	45 kg/cm ² G	a) & b) A fall of MPW pressure leads to a decrease in the purging capacity. a) 53K MPW system; PD-301 system b) 30K MPW system; PD-302, PD-303 systems	Ditto	(2)-a)-① Adjust the pressure of MPW by controlling PIC-2503 on manual. ② Check the working condition of PV-2503. ③ Check the variation of MPW to be used. ④ Confirm that PP-503 is functioning properly. ⑤ Check the pressure on PIC-2502.
	b) PIC-2701	30 kg/cm ² G	25 kg/cm ² G (Alarm)	24 kg/cm ² G			(2)-(b)-① Take the same actions as mentioned in the items (1)-a) above. ② Adjust the pressure of MPW by controlling PIC-2701 on manual. ③ Check the working condition of PV-2701.

Equipment	Factor	Operational Control Figure			Reason for Setting of Critical Figure	Method of Detection	Example of Action to be Taken upon Deviation of Control Figure from Normal One
		Normal	Allowable	Critical			
(PD-1206)	2) <u>Temperature</u> (1) Upper Limit (TIC-2706)	200°C	a) 210°C (Alarm)	b) 220°C (Design)	(1) If MPW is heated over the temperature on the process side, there is a possibility that instrument malfunctions due possibly to a flush phenomenon to be caused in the conduit pipe.	Temperature Indicator and Regulator with Alarm (TIC-2706)	(1)-a)-① Adjust the temperature of MPW by controlling TIC-2706 on manual. ② Check the working condition of TV-2706. ③ Check the temperature on TIC-2504. ④ Check the variation of an amount of MPW to be heated in PD-1206. ⑤ Check the pressure of steam fed to PD-1206. (1)-b)-① When the indication on TIC-2706 exceeds 220°C, stop feeding steam to PD-1206.
	(2) Lower Limit (TIC-2706)	200°C	190°C (Alarm)	190°C	(2) A fall of MPW pressure leads to a decrease in the purging capacity.	Ditto	(2)-① Take the same actions as mentioned in the items (1)-a)-① ~ ⑤ above. ② Check the working condition of steam trap. (ST-2201)
54. Recycle Solvent Heater (PE-403)	<u>Temperature</u> (1) Upper Limit (TIC-2403)	101°C	a) 110°C (Alarm)	b) 120°C (Design)	(1) It is specified to prevent hot water from spouting to the atmosphere due to an excessive temperature rise.	Temperature Indicator and Regulator with Alarm (TIC-2403)	(1)-a)-① Adjust the temperature of R.S. by controlling TIC-2403 on manual. ② Check the working condition of TV-2403. ③ Check the indications on FI-2407 and FI-2409 to confirm the variation of amount of R.S. (1)-b)-① Stop feeding steam to PE-403. (Close TV-2403 on manual)

Equipment	Factor	Operational Control Figure			Reason for Setting of Critical Figure	Method of Detection	Example of Action to be Taken upon Deviation of Control Figure from Normal One
		Normal	Allowable	Critical			
(PE-403)	(2)Lower Limit (TIC-2403)	101°C	95°C (Alarm)	90°C	It is specified to prevent the Stress Corrosion Cracking (SCC) from occurring on the equipment of the process system due to an increase in oxygen content dissolved in solvent following the decrease in the temperature.	Ditto	(2)-① Take the same actions as mentioned in the item (1)-a)-① through ③ above.
55.Hot Oil Heater (PF-601)	1)Hot Oil Flow Lower Limit (FICS-2601)	545t/hr	a)450t/hr (Alarm)	b)200t/hr Interlock	It is specified to prevent the heating tube in PF-601 from being overheated and damaged due to a decrease in the hot oil flow fed to PF-601.	Flow Indicator and Regulator with Alarm and Interlock (FICS-2601)	a)-① Adjust the flow rate of Hot Oil fed to PF-601 by controlling FICS-2601 on manual. ② Check to be sure that the flow control valve FV-2601 is working properly. ③ Confirm that there is not any problem with the Hot Oil Circulation Pump PP-601 A, B or C. ④ In case where the reactor temperature could not be kept at the specified one, switch the operation over to the Recycle Solvent Run. b)- ① According to the actuation of the Interlock, all main burners are to be extinguished. For the process system, immediately stop feeding CTA to PD-101 and switch the operation over to the Recycle Solvent Run in preparation for the ESD.

Equipment	Factor	Operational Control Figure			Reason for Setting of Critical Figure	Method of Detection	Example of Action to be Taken upon Deviation of Control Figure from Normal One
		Normal	Allowable	Critical			
(PF-601)	2) <u>Hot Oil Temperature</u> (1) Upper Limit (TICSK-2601)	313°C	a) 325°C (Alarm)	b) 345°C Interlock	(1) It is specified to prevent the heating tube in PF-601 from being overheated and damaged due to abnormal temperature rise of Hot Oil.	Temperature Indicator and Regulator with Alarm and Interlock (TICSK-2601)	(1)-a)-① Switch FIC-2611 (Fuel Flow) and FIC-2610 (Air Flow) over to the manual mode, and adjust the Hot Oil temperature by controlling them. ② Check the flow rate of Hot Oil fed to PF-601 (FICS-2601) and also each user. ③ Adjust the flow rate of Hot Oil fed to PE-601 to control the temperature of Hot Oil.
	(2) Lower Limit (TICSK-2601)	313°C	305°C (Alarm)	303°C	(2) It is specified to prevent the pd/c catalyst in PD-201 from being crushed due to crystallization of TA dissolved in PD-201.	Ditto	(1)-b)-① The same actions as mentioned in the item (1) Hot Oil Flow b)-① above. (2)-① Take the same actions as mentioned in the items (1)-a)-① through ③ above. ② Check the main burners for their firing conditions. ③ For the process system, immediately stop feeding CTA to PD-101, and switch the operation over to the Recycle Solvent Run.
	3) <u>Pressure of FG</u> Lower Limit (PIS-2611)	1.0 kg/cm ² G	a) 0.7 kg/cm ² G (Alarm)	b) 0.2 kg/cm ² G Interlock	It is specified to prevent the main burner from extinguishing as well as to avoid gas explosion followed by the extinction of burner due to a fall of the FG pressure.	(1) FG Pressure Indicator with Alarm and Interlock (PIS-2611)	(1)-a)-① Check the indication on FIC-2611. (Flow rate of FG) ② Check the indication on PC-2602 and the working condition of PCV-2602. ③ Confirm that pilot burners are ignited properly. (1)-b)-① FG main burners are extinguished by the actuation of the Interlock (SDV-2610A/B SHUT) ② Immediately switch the operation over to the Recycle Solvent Run.

Equipment	Factor	Operational Control Figure			Reason for Setting of Critical Figure	Method of Detection	Example of Action to be Taken upon Deviation of Control Figure from Normal One
		Normal	Allowable	Critical			
(PF-601)							③ In case where the pilot burners are extinguished by the actuation of the interlock, (SDV-2609A/B SHUT) move to the ESD.
56.Hot Oil Buffer Drum (PD-601)	1) <u>Pressure</u> (1) Upper Limit (PC-2608)	0.4 kg/cm ² G		1.8 kg/cm ² G (Design)	(1) It is specified to prevent the drum from being overpressurized because PD-601 is provided with 7N seal system to avoid the degradation of Hot Oil in contact with air.	Pressure Indicator and Regulator (Field) (PC-2608)	(1)-① Confirm that the pressure control valves PCV-2608A/B are working properly. ② Check the indication on PC-2608 by comparing it with that on PG-2604. ③ Close the block valve of PCV-2608A. ④ Check the indication on FI-2608 to be sure that the flow rate of vent gas is in the range of normal condition. ⑤ When the pressure on PC-2608 rises to 1.8 kg /cm ² G, SV-2601 actuates and releases the pressure.
	(2) Lower Limit (PC-2608)	0.4 kg/cm ² G		0 kg/cm ² G	(2) It is specified to prevent PP-601 from getting into the cavitation phenomenon.	Ditto	(2)-① Confirm that the pressure control valves PCV-2608A/B are working properly. ② Check the indication on PC-2608 by comparing it with that on PG-2604. ③ Check to be sure that there is not any plugging around the vent line, pressure indicator and the lead pipe, etc. ④ Check the pressure of 7N fed to PC-2608, which is indicated on PG-3131.
	2) <u>Vent Gas Flow</u> Upper Limit (FI-2608)	0Nm ³ /hr	50Nm ³ /hr (Alarm)	100Nm ³ /hr	It is specified to promptly detect an inclusion of water to Hot Oil system.	Flow Indicator with Alarm (FI-2608)	① Confirm that PC-2608 is functioning properly. ② Check the opening degree of PCV-2608B.

Equipment	Factor	Operational Control Figure			Reason for Setting of Critical Figure	Method of Detection	Example of Action to be Taken upon Deviation of Control Figure from Normal One
		Normal	Allowable	Critical			
(PD-601)							③ In case where the indication on FI-2608 doesn't fall and rises further even with the block valve of PCV-2608A being closed, immediately move to the ESD because it is presumed that water is entering Hot Oil system from the process side.
	3) <u>Liquid Level</u> (1) Upper Limit (LI-2601)	55%	70% (Alarm)	90%	(1) It is specified to prevent Hot Oil in PD-601 from overflowing.	Level Indicator with Alarm (LI-2601)	(1)-① Check the indication on LI-2601 by comparing it with that on LG-2601. ② Transfer H.O. in PD-601 to PTK-600 to reduce the level.
	(2) Lower Limit (LI-2601)	55%	52% (Alarm)	50%	(2) It is specified to prevent PP-601 from getting into the cavitation phenomenon.	Ditto	(2)-① Check the indication on LI-2601 by comparing it with that on LG-2601. ② Make up for PD-601 with Hot Oil by sending it from PTK-600 to PD-601. ③ Check the Hot Oil line to be sure that there is no leakage of H.O. in the field.
57. Hot Oil Tank (PTK-600)	1) <u>Pressure</u> (1) Upper Limit (PC-2600)	25mmAq	a) 30mmAq (Seal Pot)	b) 70mmAq (Relief Hutch)	(1) It is specified to prevent PTK-600 from being damaged due to an overpressurization because it is designed by the pressure of 100mmAq.	Pressure Regulator (PC-2600)	(1)-a)-① Confirm that PC-2600 is working properly. ② Check to be sure that the seal leg of PZ-600 is correct. ③ Check to be sure that any safety valve doesn't actuate for the Hot Oil system. (1)-b)-① In case where the relief hutch actuated, switch the operation over to the Recycle Solvent Run and then, move to the ESD.

Equipment	Factor	Operational Control Figure			Reason for Setting of Critical Figure	Method of Detection	Example of Action to be Taken upon Deviation of Control Figure from Normal One
		Normal	Allowable	Critical			
(PTK-600)	(2)Lower Limit (PC-2600)	25mmAq	0mmAq	-25mmAq (Breather Valve)	(2)It is specified to prevent PTK-600 from being damaged due to a negative pressure.	Ditto	(2)- ① Confirm that PC-2600 is working properly. ② Check to be sure that the seal leg of PZ-600 is correct. ③ Check the pressure of 7N fed to PC-2600, which is indicated on PG-2602.
	2)Liquid Level Upper Limit (LG-2600)			85m ³ (4.8m)	It is specified to prevent Hot Oil from overflowing due to the over-charge of it.	Magnetic-Type Level Indicator (LG-2600)	① Stop receiving H.O. if it is being received into PTK-600. ② Stop recovering H.O. from PD-603. ③ Check to be sure that there is not any H.O. leaked from the outlet of PP-601 to PTK-600. ④ Check to be sure that any safety valve doesn't actuate for the Hot Oil system.
58.Residue Slurry Drum (PD-901)	Liquid Level (1)Upper Limit (LIC-2911)	65%	85% (Alarm)	90%	(1)It is specified to prevent FDQ from overflowing due to charging an excessive amount of FDQ. Note; The level of 100% on LIC-2911 is equivalent to the position just below the SG.	Level Indicator and Regulator with Alarm (LIC-2911)	(1)-① Check the indication on LIC-2911 by observing the actual level through SG. ② Check an amount of FDQ charged through FIC-2911 to PD-901 and the working condition of FV-2911. ③ If there is any problem with FV-2911, adjust the level by-passing it. ④ Check to be sure that residue separated in PM-901 doesn't have an excessive amount of water.
	(2)Lower Limit (LIC-2911)	65%	25% (Alarm)	20%	(2)It is specified to prevent PP-902 from getting into the cavitation phenomenon as well as to avoid a clogging due to variation of slurry concentration.	Ditto	(2)-① Take the same actions as mentioned in the items (1)-①~③ above. ② Check the pressure of FDQ charged to PD-901 from TD-302, which is indicated on PG-1302(TA unit).

Equipment	Factor	Operational Control Figure			Reason for Setting of Critical Figure	Method of Detection	Example of Action to be Taken upon Deviation of Control Figure from Normal One
		Normal	Allowable	Critical			
(PD-901)							③ Confirm that TP-302 is pumping up properly.
59. H.O. Underground Drum (PD-603)	<u>Liquid Level</u> (1) Upper Limit (LIK-2603) (2) Lower Limit (LIK-2603)		80% (Alarm)	80%	(1) It is specified to prevent H.O. from overflowing. (2) It is specified to prevent PP-603 from getting in the dry-run.	Level Indicator with Alarm (LIK-2603) Ditto	(1)-① Confirm that Hot Oil is not charged through the funnel and drain line to PD-603. ② Start PP-603 to send H.O. to PTK-600. (2)-When the level on LIK-2603 falls down to 10%, PP-603 is halted automatically by the actuation of the Interlock.
60. Fan FDF (PC-601) IDF (PC-602)	<u>Ampere</u> Upper Limit (IJI-2611) (IJI-2612)	40A 85A	a) 45A (Alarm) 89A (Alarm)	b) 49A (Rated) 90A (Rated)	They are specified to prevent PC-601 and PC-602 from being damaged due to overcurrent.	Ampere Indicator with Alarm (IJI-2611) (IJI-2612) Stop Alarm with Interlock	a)-① Adjust the amount of draft. ② Check the fan to be sure that there is not any problem with it. If any problem, switch the operation over to the Recycle Solvent Run. b)-① According to the stoppage of fan, all main and pilot burners are extinguished, and the operation mode of PF-601 is switched over to the natural draft operation by the actuation of the Interlock. ② Switch the operation over to the Recycle Solvent Run stopping feeding CTA to PD-101, and move to the ESD.
61. Air Preheater (PE-605)	1) <u>Flue Gas</u> Temperature (TJI-2623)	180~200°C	160°C (Alarm)	132°C	It is specified to prevent the convection section of PF-601 from being corroded by the sulfuric acid condensed on the pipe due to a fall of temperature for the flue gas.	Temperature Indicator with Alarm (TJI-2623)	① Adjust the temperature of flue gas by controlling the Cold-By-Pass Valve installed on the PE-605. Note; Flue gas temperature is specified by the vendor.

Equipment	Factor	Operational Control Figure			Reason for Setting of Critical Figure	Method of Detection	Example of Action to be Taken upon Deviation of Control Figure from Normal One
		Normal	Allowable	Critical			
62. Recovered TA Slurry Drum (PD-901)	<u>Liquid Level</u> Upper Limit (LIC-2911)	50%	60% (Alarm)	90%	It is specified to prevent the slurry from overflowing as well as to avoid a possibility of plugging the bottom of PM-902 with PTA residue.	Level Indicator and Regulator with Alarm (LIC-2911)	① Confirm that an excessive amount of PTA residue is not discharged through cute to PM-901 from PD-901. ② Check to be sure that there is not any place clogged with PTA residue on the suction and delivery lines of PP-902 A/B. ③ Check the working condition of FV-2911 and adjust the level by controlling FV-2911 on manual, if necessary. ④ Switch PP-902 over to the stand-by side. ⑤ Switch PTA residue from PD-901 to PZ-903, if necessary.
63. Agitator (PJ-901)	<u>Ampere</u> Upper Limit (Ampere Meter in the field)	6A	7.5A (Alarm)	7.5A (Rated)	It is specified to prevent the motor from being damaged due to the overload.	Ampere Meter with stop Alarm	① Check the indication on LIC-2911. ② Check to be sure that there is not any problem with the motor and reducer. If it's judged impossible to continue operation based on the facts found in the action above, stop feeding PTA residue to PD-901.
64. Recoverd TA Slurry Filter (PM-903A/B)	<u>Differential pressure</u> Upper Limit (PDI-2903)	0.1 kg/cm ² G	1.0 kg/cm ² G (Alarm)	1.2 kg/cm ² G	It is specified to prevent the filter element from being damaged due to a high differential pressure.	Differential Pressure Indicator with Alarm (PDI-2903)	① Switch the filter over to the stand-by side and clean it.